MEGO-C: A Novel Interpretable Machine Learning Architecture for Fusing Satellite and In Situ Data Into a Submesoscale-Resolving 3-D Chlorophyll Atlas

Yanfeng Wen, Cédric Jamet, Peng Chen, Kun Shi, and Delu Pan

***Abstract*—Observing the ocean's biological carbon pump at its critical meso- and submesoscales remains a significant challenge, leading to major uncertainties in global carbon cycle estimates. This challenge stems from a fundamental data disparity: while satellite ocean color sensors provide unparalleled global coverage of the sea surface, they cannot penetrate the ocean's interior. Conversely, in situ profilers like BGC-Argo measure the vertical dimension but are too sparse to resolve fine-scale horizontal variability. Existing 3-D chlorophyll-*a* (Chl-*a*) products, whether based on statistical models or numerical reanalysis, are often too coarse to resolve these key dynamics. To address this longstanding scale-mismatch problem, we introduce a multimodal ensemble for global-ocean chlorophyll-a (MEGO-C), a novel machine learning architecture specifically engineered for Earth science data fusion. It employs an interpretable, heterogeneous stacked ensemble to synergistically fuse 144 936 Biogeochemical-Argo (BGC-Argo) profiles with a suite of multisensor satellite observations. This framework produces the first-ever global, daily, submesoscale-resolving (~4 km) 3-D Chl-*a* atlas. Validation demonstrates its breakthrough performance, with prediction errors less than half of those from benchmark Copernicus Marine Environment Monitoring Service (CMEMS) products and an order-of-magnitude finer spatial resolution (~4 versus ~25 km). Crucially, our eddy case studies based on this high-resolution dataset show that coarse-resolution products can substantially underestimate area-integrated Chl-*a* ($\sum$Chl·dA; biomass proxy) within Chl-*a*-based eddy-like coherent features by ~25% and miss peak concentrations by over 50%. This finding suggests the biogeochemical impact of eddies is substantially undervalued in global assessments, a result with important implications for revising carbon cycle estimates. Shapley additive explanations (SHAP) analysis confirms the model's physical consistency, moving beyond a "black-box" approach, while scale-selective filtering verifies its ability to resolve submesoscale filaments and sharpen eddy variability. This work provides a crucial new observational benchmark for validating the biogeochemical components of Earth system models and presents a globally applicable, effective, and transparent approach for synergistic data fusion in remote sensing and oceanography.**

***Index Terms*—3-D chlorophyll-a reconstruction, biogeochemical-argo (BGC-Argo), ensemble machine learning, satellite ocean color, Shapley additive explanations (SHAP) interpretability, submesoscale variability.**

I. Introduction

ACCURATELY quantifying the ocean's role in the global climate system, particularly its capacity as a carbon sink, stands as a central challenge in global change biology. A primary source of uncertainty stems from the widespread omission of meso- and submesoscale processes (1–100 km)—such as eddies, fronts, and filaments—in global models and data products [1], [2], [3], [4], [5]. These features are now recognized as biogeochemical hotspots that fundamentally shape marine ecosystems by driving vertical nutrient transport, regulating primary production, and influencing deep-ocean carbon export [5], [6], [7], [8]. The misrepresentation or complete neglect of these fine-scale dynamics introduces substantial uncertainty into our estimates of the ocean's carbon cycle and its response to environmental change, thereby limiting our ability to predict the future of marine biodiversity and biogeochemistry. This creates a critical "resolution gap" between the scale of key biogeochemical processes and the scale of our observations.

For decades, satellite remote sensing has revolutionized oceanography by providing global, daily views of surface chlorophyll-*a* (*Chl-a*), a key proxy for phytoplankton biomass [9]. Yet, this 2-D perspective leaves the vast majority of the ocean's productive volume—the expansive euphotic zone—unobserved from space. This limitation is critical, as the vertical structure of phytoplankton communities dictates the efficiency of the biological pump and the functioning of pelagic food webs. Concurrently, the Biogeochemical-Argo (BGC-Argo) program provides invalu-

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able in situ vertical profiles, but the sparse distribution of its floats cannot resolve the high-resolution horizontal variability needed to study these dynamic features [10]. This fundamental data disparity—between sparse, point-based, vertically resolved in situ measurements and dense, spatially continuous, surface-only satellite observations—constitutes a classic and longstanding data fusion challenge for accurately assessing the global distribution of phytoplankton, the foundation of most marine food webs [11].

Previous efforts to reconstruct subsurface *Chl-a* from surface data have employed a wide range of approaches. These fall broadly into two categories: physics-based biogeochemical models and data-driven statistical methods. While biogeochemical models provide a dynamically consistent 3-D view, they often suffer from significant regional biases [12], [13] and their coarse resolution (typically $\geq 0.25°$) prevents them from resolving the fine-scale features of interest [5], [14]. Data-driven methods, ranging from simple empirical regressions [15], [16] to more sophisticated machine learning techniques like artificial neural networks (ANNs) [15], [17] and ensemble-based models [18], [19], [20], have shown notable success. However, these methods often lack the generalizability to perform consistently across diverse oceanic regimes, and their "black-box" nature can limit scientific trust and understanding of the underlying physical drivers [21], [22], hindering their full potential for scientific discovery.

To address these limitations, we developed a novel methodological solution: a new, high-resolution global 3-D Chl-*a* product, multimodal ensemble for global-ocean chlorophyll-*a* (MEGO-C). This article presents this new dataset and the innovative framework behind its creation. Leveraging a state-of-the-art heterogeneous ensemble learning framework, we synergistically fuse the vertical completeness of the BGC-Argo network with the horizontal coverage of multisensor satellite observations. The primary contributions of this work are threefold.

1) *Methodological Innovation:* We introduce MEGO-C, a novel and interpretable heterogeneous ensemble learning architecture specifically designed for robust 3-D geophysical field reconstruction, which overcomes the limitations of single-model and "black-box" approaches.

2) *Product Breakthrough:* We present and validate the first global, daily, submesoscale-resolving 3-D Chl-a dataset, demonstrating a marked improvement in accuracy and a revolutionary leap in spatial resolution over existing benchmark products.

3) *New Scientific Insight:* Using this unprecedented dataset, we provide direct observational evidence that previous coarse-resolution approaches have led to a substantial underestimation of phytoplankton biomass in these scale-selective, Chl-a-defined biogeochemical hotspots, with critical implications for global carbon budget estimates.

II. Data

Our methodology relies on fusing in situ vertical profiles with gridded surface and 3-D datasets. All data sources, their key variables, and resolutions are summarized in Table I.

A. BGC-Argo Profiles

We used all available quality-controlled BGC-Argo profiles from October 2005 to December 2024, obtained from the Euro-Argo data portal. We focused on the upper 200 m, extracting profiles of temperature (*TEMP*), salinity (*PSAL*), and HPLC-calibrated *Chl-a* (*CHLA_ADJUSTED*) [3]. Raw fluorescence was corrected for nonphotochemical quenching (*NPQ*) and instrument bias following established protocols [23]. This resulted in a core dataset of 144 936 high-quality profiles for model training and validation.

B. Satellite and Reanalysis Data

We compiled a comprehensive set of environmental drivers from satellite remote sensing and reanalysis products (Table I). These include.

1) *Ocean Optics:* Daily, 4 km multisensor remote-sensing reflectance (Rrs at 412, 443, 490, 555, 670 nm), diffuse attenuation coefficient (*Kd(490)*), and other optical variables from the Copernicus-GlobColour product (DOI: 10.48670/moi-00280) [24].

2) *Sea Level and Winds:* Daily sea-level anomaly (*SLA*) data were obtained from the Copernicus C3S-DUACS product (DOI: 10.48670/moi-00145) [25]. Surface wind fields were derived from the CCMP Version 3.1 dataset provided by Remote Sensing Systems, including both 6-h global vector winds and monthly wind anomalies to capture dynamic ocean processes [26].

3) *Physical Oceanography:* Daily mixed-layer depth (*mlotst*), 3-D temperature (*thetao*), and 3-D salinity (*so*) from the 1°/12° GLORYS12V1 global physical reanalysis (DOI: 10.48670/moi-00021) [27]. Among these, *mlotst* was used as a predictive feature during model training to provide the 3-D physical context for *Chl-a* reconstruction, whereas *thetao* and *so* were used exclusively during postprocessing to support the generation of the final 3-D *Chl-a* fields.

C. Reference Datasets for Comparison

To benchmark the performance of MEGO-C, we selected three widely used global 3-D chlorophyll-*a* (*Chl-a*) datasets provided by the Copernicus Marine Environment Monitoring Service (CMEMS). These products were chosen because they represent the three primary categories of available global 3-D Chl-a data: a statistical observation-based product, a near-real-time biogeochemical model forecast, and a reanalysis product, thus ensuring a comprehensive and fair comparison. These include.

1) A statistical observation-based product (CMEMS MULTIOBS (Dataset 1); DOI: 10.48670/moi-00046), which integrates satellite and in situ observations via a neural network approach [15].

2) A near-real-time biogeochemical model forecast (CMEMS BGC Forecast (Dataset 2); Global Ocean Biogeochemistry Analysis and Forecast; DOI: 10.48670/moi-00015) [28], [29].

3) A reanalysis product (CMEMS BGC Reanalysis (Dataset 3); Global Ocean Biogeochemistry Hindcast; DOI: 10.48670/moi-00019) [30].

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TABLE I
SUMMARY OF OBSERVATIONAL, SATELLITE, AND REANALYSIS DATASETS USED IN THIS STUDY. KEY VARIABLES, ORIGINAL SPATIAL AND TEMPORAL RESOLUTIONS, AND DATA SOURCES ARE PROVIDED FOR EACH PRODUCT

Dataset, Product ID / version	Sensor / data type	Key variables used	Resolution (space × time)
BGC-Argo profiles	In-situ profiling floats	*TEMP, PSAL, CHLA, PRES*	Point; ≈ 1 m vertical × < 10 d
Copernicus-GlobColour BGC L3	Multi-sensor satellite fusion	*Rrs(λ), Kd(490), ZSD, SPM*	4 km × daily
C3S-DUACS sea-level L4	Gridded altimetry	*SLA*	0.25° (~28 km) × daily
CCMP vector winds L4 / L4.5	Blended scatterometer & radiometer	*uwnd, vwind, u_anom, v_anom*	0.25° × 6 h / monthly
GLORYS12V1 reanalysis	NEMO model with data assimilation	*mlotst, thetao, so*	1⁄12° (~8 km) × daily
CMEMS MULTIOBS (Dataset 1)	In-situ & Satellite fusion	*Chl-a*	0.25° × Weekly
CMEMS BGC Forecast (Dataset 2)	Numerical model	*Chl-a*	0.25° × Daily
CMEMS BGC Reanalysis (Dataset 3)	Numerical model	*Chl-a*	0.25°× Daily

Detailed characteristics of these datasets, including spatial resolution, temporal frequency, and data sources, are summarized in Table I.

III. MEGO-C ENSEMBLE FRAMEWORK

The workflow for developing MEGO-C is illustrated in Fig. 1. It comprises five main steps: 1) data acquisition and collocation; 2) preprocessing and feature engineering; 3) model training and selection; 4) global 3-D reconstruction; and 5) product validation and interpretation.

A second 3-D predictor cube is assembled for inversion (see Section III-C). Because some fields differ from those used for training, a mild statistical adjustment is applied, after which the same preprocessing and feature-engineering steps are reused. The MEGO-C ensemble is then migrated to this cube to produce a global-ocean, multiyear, daily 3-D chlorophyll-*a* field at 0.0416° resolution. Each stage—data preparation, model training, migration, and reconstruction—is documented together with the corresponding validation checks to ensure robustness and reproducibility.

A. Data Preprocessing and Feature Engineering

All BGC-Argo profiles were collocated with the nearest-neighbor gridded data in space and time. A strict quality control procedure was applied, removing outliers based on physical ranges and uncertainty thresholds (see Table II). The final matched dataset contains 144 936 profiles, with broad global, seasonal, and vertical coverage (see Fig. 2).

For each BGC-Argo profile, the four core variables—temperature (T = *TEMP*), salinity (S = *PSAL*), in vivo chlorophyll-*a* fluorescence (*CHLA*), and pressure

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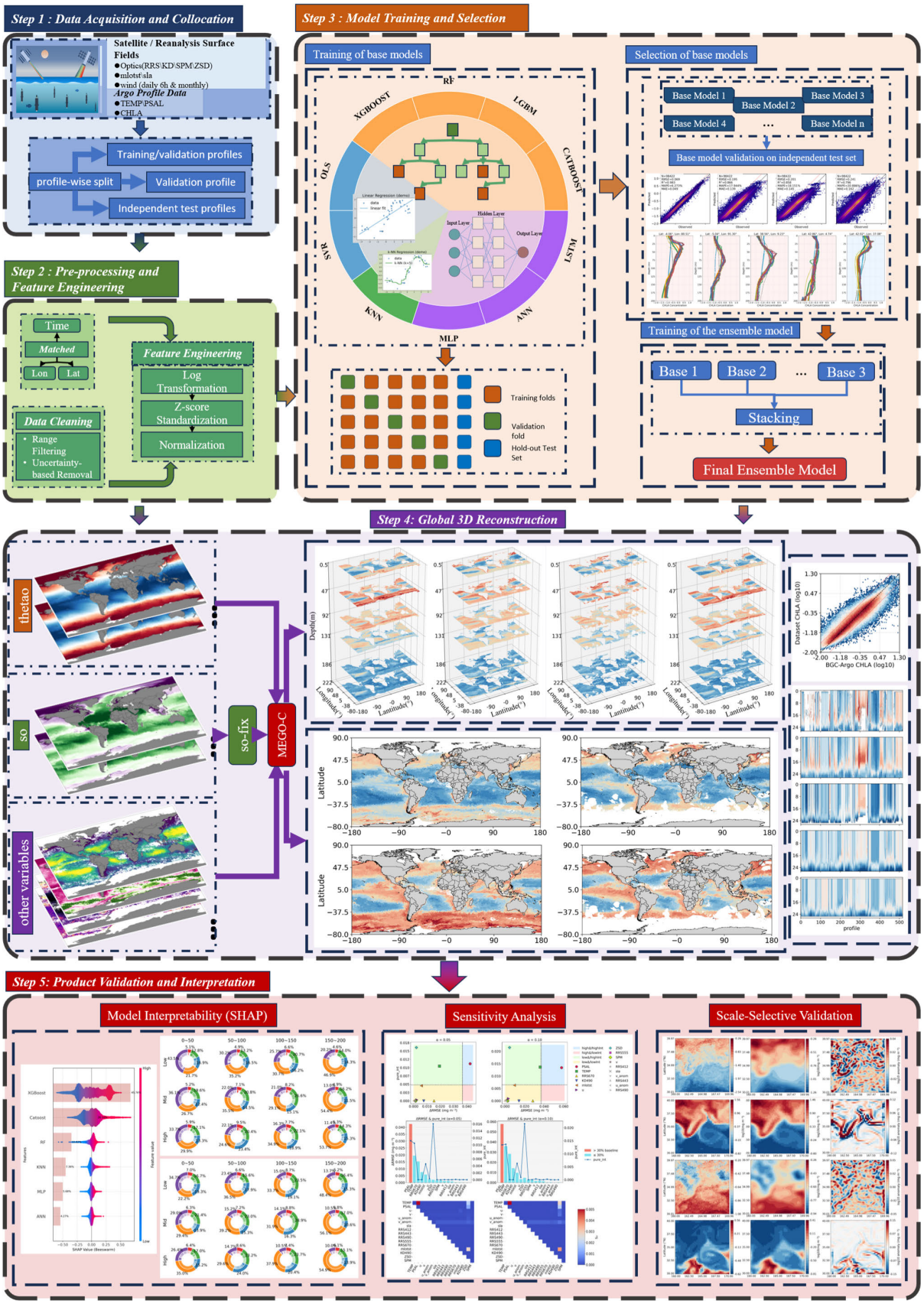


Fig. 1. Schematic workflow of the MEGO-C framework. Step 1: BGC-Argo profiles are collocated with satellite and reanalysis data. Step 2: The matched data undergo rigorous preprocessing and feature engineering. Step 3: A diverse suite of base learners is trained and evaluated, with the top performers selected to build a stacked ensemble model (MEGO-C). Step 4: The trained MEGO-C model is deployed on a global, high-resolution 3-D predictor dataset to reconstruct the Chl-a field. Step 5: The final product is rigorously validated and interpreted using SHAP analysis, sensitivity testing, and multiscale eddy detection.

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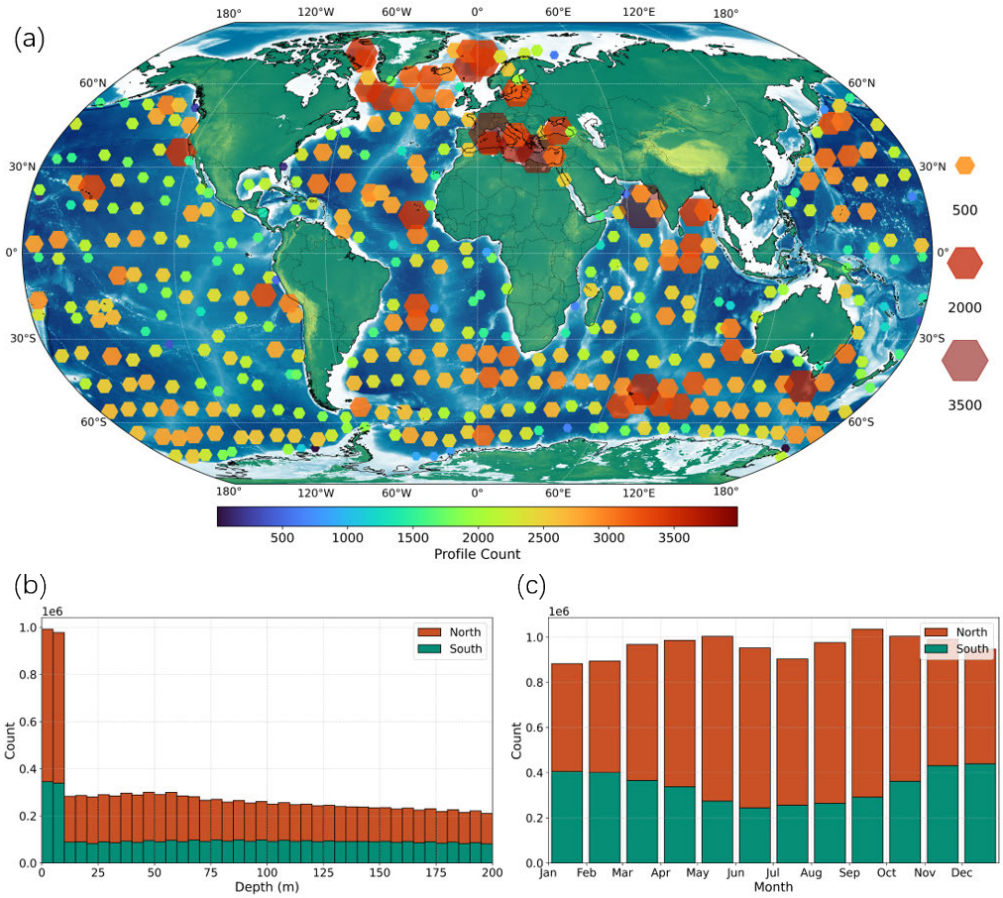


Fig. 2. Global distribution and sampling density of the 144 936 quality-controlled BGC-Argo profiles used for model development. (a) Spatial distribution, with hexagons colored by profile count per 5° × 5° bin, showing globally representative sampling. (b) Vertical distribution of observations, demonstrating dense coverage throughout the upper 200 m. (c) Monthly distribution of observations, indicating balanced seasonal coverage. Data are split between the Northern (orange) and Southern (teal) Hemispheres.

TABLE II

VARIABLE-SPECIFIC PLAUSIBILITY RANGES AND UNCERTAINTY THRESHOLDS ADOPTED FOR QUALITY-CONTROL SCREENING

Variables	Accepted range/threshold	Units
_uncertainty (all)	≤ 80	*% (relative)*
RRS (412, 443, 490, 555, 670)	0 – 0.07	sr^{-1}
TEMP	–2 – 35	*°C*
PSAL	0 – 42	*PSU*
DEPTH	≥ 0	*m*
mlotst	0 – 1500	*m*
SLA	–2 – 2	*m*
uwnd, vwind (6-hour) *u(anom), v(anom) (monthly)*	-10-10	$m*s^{-1}$
CHLA	0.01 – 20	$mg*m^{-3}$
KD(490)	0.01 – 1	m^{-1}
SPM	0 – 50	$g*m^{-3}$
ZSD	0.5 – 100	*m*

(*PRES*)—were inspected within the 0–200 m layer. A profile was retained only if all four variables contained at least one "good-quality" observation in this depth range; otherwise, it was discarded. Remaining profiles were despiked with a three-point moving average, and the raw fluorescence was recalibrated to laboratory units with the HPLC-constrained relationship [23], [31], [32]

$$CHLA_{model} = \frac{CHLA_{Bgc-Argo} - 0.019}{2.32} \tag{1}$$

thereby removing instrument bias. Sea PRES (*p, dbar*) was then converted to geometric depth (*z, m*) using the

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TABLE III
FEATURE ENGINEERING TECHNIQUES UTILIZED IN MEGO-C

Variables	Transformation
CHLA	Log → Normalize
KD(490), RRS(490), RRS(555), RRS(670), SPM, ZSD, mlotst	Log → Z-score → Normalize
RRS(412), RRS(443), TEMP, PSAL, sla, uwnd(u), vwind(v), uanom, vanom	Z-score → Normalize
LATITUDE (φ)	Sin → Normalize
LONGITUDE (v), JULD (θ)	Sin & Cos → Normalize
DEPTH	Clip → Normalize

TEOS-10 routine *gsw_z_from_p* (GSW v3.6.15) [33], [34], which accounts for the latitude dependence of gravity and provides a common vertical axis for all subsequent analyses.

Non-Argo variables were collocated to each profile by exact date for daily products, by the nearest six-hour slot for CCMP winds, or by calendar month for monthly anomalies; in the horizontal dimension, the nearest grid cell to the float's latitude–longitude position was selected. After collocation, gross outliers were removed using physically based validity ranges (see Table II); for satellite optical products [*RRS, KD(490), ZSD, SPM*], only observations with relative uncertainties below 20% were kept. After spatiotemporal collocation, we obtained 144 936 valid Argo satellite match-ups, which constitute the complete training dataset; their coverage is illustrated in Fig. 2.

Fig. 2(a) shows that the profiles are distributed across all major ocean basins and marginal seas, providing a globally representative sampling. Fig. 2(b) indicates that data density is highest in the upper 10 m and remains nearly uniform from 10 to 200 m, a pattern that supports superior model skill at the surface while ensuring robust performance throughout the water column. Fig. 2(c) shows that, after quality control, the sampling frequency is broadly uniform throughout the 12-month cycle; seasonal bias is therefore negligible, and the model should generalize well across seasons. Spatially, the Northern Hemisphere still contains slightly more profiles and quality-controlled observations than the Southern Hemisphere. This imbalance reflects the deployment pattern of the BGC-Argo fleet: from the program's inception in 2000–2009, the region south of 30 °S consistently carried several hundred fewer floats than the target array [2], [4]. In addition, the severe sea-ice and low-temperature conditions of the Southern Ocean lead to extra profile rejections during delayed-mode quality control, so the number of usable observations in the south is correspondingly lower.

To prepare the data for machine learning, we engineered a set of 22 input features. *Chl-a* and other skewed variables were log-transformed to approximate a normal distribution. All features were standardized (*Z-score*) and normalized to a [−1, 1] range to ensure numerical stability. To handle periodicity, geographic longitude and day-of-year were encoded using sine and cosine transformations. These steps are crucial for optimizing model performance.

Specifically, to capture both the symmetric similarities and potential differences between the Northern and Southern Hemispheres, we applied a sine transform to latitude (sin φ) during feature engineering, which maps equivalent latitude bands in the two hemispheres to emphasize their symmetry while allowing for the capture of possible interhemispheric disparities. To avoid discontinuities at the 0°/180° meridian, longitude (v) was encoded using both its sine and cosine components as input features. This transformation enables the model to account for the circular nature of geographic coordinates. Similarly, *JULD* (i.e., the sampling date) was first normalized to its relative position within the year and then converted to a radian value θ; the sine and cosine of θ were also used as input features, allowing the model to effectively capture seasonal periodicity and avoid discontinuities at the year boundary. These transformations are illustrated in (2), where N is the number of days in the year. *DEPTH* was linearly scaled to the [−1, 1] range to match the magnitude of the other predictors. All feature-processing steps (including those applied to the *CHLA* target) are summarized in Table III; they aim to render each feature's marginal distribution as close as possible to a normal or uniform shape while preserving the underlying data structure, thus ensuring compatibility with a broad class of modeling approaches. The Z-score standardization transforms each observation x by subtracting the sample mean μ and dividing by the sample standard deviation σ, yielding a new variable with zero mean and unit variance, as given in (3)

$$\theta = 2\pi * \frac{JULD}{N} \tag{2}$$

$$x' = \frac{x - \mu}{\sigma} \tag{3}$$

$$y = log\,(x + \varepsilon)\,. \tag{4}$$

As detailed in (4), we perform the logarithmic transformation with a small offset $\varepsilon = 10^{-5}$ added to each raw value, thereby ensuring numerical stability by preventing the input to the logarithm from reaching zero. Finally, all variables are normalized to the [−1, 1] interval to balance their numerical scales. Accordingly, these transformations yielded a set of 22 features used as input variables for model fitting.

B. Ensemble Model Development

We adopted a stacked ensemble strategy to harness the complementary strengths of different algorithmic approaches.

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Level-1 (Base Learners): We evaluated a diverse suite of ten regression algorithms from four families chosen for their distinct ways of modeling data relationships: tree ensembles [Random Forest (RF), eXtreme Gradient Boosting (XGBoost), LightGBM, and Categorical Boost (CatBoost)], neural networks [feed-forward ANN, multilayer perceptron (MLP), and long short-term memory (LSTM) network], kernel methods [ordinary-least-squares (OLS) and support-vector regression (SVR)], and distance-based learners [k-nearest neighbors (KNNs)].

Level-2 (Meta-Learner): The six best-performing base models were selected based on their performance (R^2, RMSE) in fivefold cross-validation. The out-of-fold (OOF) predictions from these models were then used as input features to train a final XGBoost meta-learner. This stacking approach creates the final MEGO-C model, which synthesizes the strengths of its constituent learners to achieve higher accuracy and better generalization than any single model could alone. This stacked architecture is not only optimized for *Chl-a* reconstruction but is also designed to be a transferable framework for other geophysical parameters where similar data fusion challenges exist.

The complete dataset was divided profile-wise, with 85% used as a development set and 15% reserved as an independent hold-out test set for an unbiased assessment of final predictive skill. In practice, this split was implemented group-wise by float/platform ID, such that all profiles from the same float were assigned to a single split (development versus hold-out), and the split was constructed to achieve an approximate 85%/15% ratio in profile counts. To strictly prevent data leakage in stacking while keeping the computational cost tractable, we adopted a staged training scheme given the sufficient sample size. Specifically, the 85% development set was further split into two disjoint subsets at an approximately 4:1 ratio, corresponding to 68% versus 17% of the total dataset: a base-learner training subset (68%) and a stacking subset (17%) (i.e., $0.85 \times 0.8 \approx 0.68$ and $0.85 \times 0.2 \approx 0.17$ of the full dataset).

In Level-1, ten candidate base learners were tuned and evaluated on the 68% subset using fivefold cross-validation with grid search, and the six best-performing models (XGBoost, CatBoost, RF, KNN, ANN, MLP) were selected. All cross-validation folds were constructed by float ID to avoid platform-level leakage. In Level-2, the selected base learners trained on the 68% subset were used to generate prediction features for the 17% stacking subset; the XGBoost meta-learner was then tuned on this stacking subset using the same fivefold cross-validation and grid search procedure. After fixing the model family and hyperparameters, we performed a final refit for deployment: all selected base learners were retrained on the entire 85% development set, and the meta-learner was trained on OOF predictions generated on this same 85% development set via fivefold cross-validation (i.e., each sample's Level-1 features are produced by models that did not use that sample for training). During inference (including evaluation on the 15% hold-out test set), Level-1 predictions from the refit base learners are fed into the trained meta-learner to produce the final MEGO-C estimates. Throughout the process, the independent holdout validation set—kept separate from the training data—was used to evaluate the performance of both each trained base learner and the overall ensemble model MEGO-C. This profile-wise split ensures that all data points from a single Argo profile belong exclusively to either the training or the test set, preventing any potential data leakage and guaranteeing a truly independent validation. Both subsets were constructed to retain representative coverage of the major oceanic regimes—low-latitude oligotrophic gyres, high-latitude nutrient-rich waters, and optically complex coastal zones—so that model evaluation would reflect the full range of environmental conditions. Fivefold cross-validation was then performed within the training subset; because each fold contained a similar mix of regimes, this procedure provided a robust estimate of goodness-of-fit and stability across diverse ocean settings.

C. Global 3-D Reconstruction

To obtain an operational global 3-D chlorophyll-*a* archive, we first removed the distributional mismatch between training and production predictors. The only variable that changed between phases—the GLORYS12V1 temperature–salinity (T/S) field—was compared with the BGC-Argo T/S profiles (2005–2024) through R^2 and root-mean-square error (RMSE) diagnostics. Guided by these statistics, we trained an XGBoost bias-correction model that takes the raw GLORYS12V1 values and their geo-temporal coordinates as inputs and returns collocated BGC-Argo measurements, thereby aligning the reanalysis mean, variance, and vertical gradients with the in situ data. The correction model used latitude, longitude, day of year, and depth, alongside the original GLORYS12V1 T/S values, to predict the bias-corrected values.

The bias-corrected T/S cube, merged with all other satellite and reanalysis drivers and uniformly upsampled to 1°/24° resolution to match the model output scale, was passed to the calibrated MEGO-C ensemble to create a daily global *Chl-a* product (0.0416° horizontal resolution; 27 depth levels from 0 to 220 m) for 2010–2024. An identical pipeline applied to monthly inputs generated a companion monthly-mean dataset. Because both products follow the original preprocessing and feature-engineering workflow, they inherit the model's calibrated skill without distributional drift.

To demonstrate scientific reliability, we collocated MEGO-C profiles with independent BGC-Argo observations and benchmarked them against three Copernicus 3-D products—*BIOGLO PISCES*, *GREEN REANALYSIS BIO*, and *MULTIOBS_GLO_BIO_BGC_3-D_REP_015_010*—over July 2021 to December 2024 (see Table I). A ±3.5 day window accommodated the weekly MULTIOBS output, and all sources were interpolated onto the common 27-level grid for profile-wise visual and statistical comparison.

D. Validation and Interpretation Strategy

To ensure scientific credibility and user trust, our validation-and-interpretation framework couples quantitative skill assessment with targeted diagnostics designed to open the "black box" of the model. Overall accuracy is quantified with the

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TABLE IV
CATEGORISATION OF FEATURE GROUPS FOR MEGO-C MODEL INPUTS

Group	Members
Depth	*DEPTH*
Optical	*RRS(412,443,490,555,670), KD(490), ZSD, SPM*
Water intrinsic	*TEMP, PSAL, mlotst*
Dynamics	*u, v, uanom , vanom , SLA*
Spacetime	*sin-LAT, sin/cos-LON, sin/cos-JULD*

coefficient of determination (R^2), RMSE, and mean absolute percentage error (MAPE) on log-transformed Chl-a. Crucially, we use Shapley additive explanations (SHAP) to unveil the contributions of each base learner and physical driver [22]; Monte-Carlo perturbations to test robustness by ranking the most critical inputs [35], [36], [37]; and an FFT-based FlowSieve filter to verify that the product retains meso- to submesoscale structure [1], [38]. Collectively, these elements deliver a coherent and comprehensive picture of MEGO-C's performance and physical credibility.

1) Performance Metrics: For model evaluation, all algorithms were benchmarked with three complementary metrics—RMSE, MAPE, and R^2—using fivefold cross-validation and an independent profile-based validation set (15% of the profiles). The formulas for the quality metrics used are presented in (5)–(7), where y_i is the observed value for sample i; $\hat{y}_i$ is the predicted value for sample i; $\bar{y}$ is the mean of all observed values; n is the total number of samples

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2} \tag{5}$$

$$RMSE = \sqrt{\frac{1}{n} * \sum_{i=1}^{n} (y_i - \hat{y}_i)^2} \tag{6}$$

$$MAPE = \frac{100\%}{n} * \sum_{i=1}^{n} \frac{y_i - \hat{y}_i}{y_i}. \tag{7}$$

2) Model Interpretability (SHAP): To reveal the drivers of MEGO-C's predictions and verify their reliability, we applied a SHAP-based attribution analysis to the ensemble model. SHAP extends game-theoretic Shapley values to machine learning by treating each input feature as a "player," the model prediction as the "pay-off," and the Shapley value as that feature's marginal contribution relative to the mean prediction [22], [39], [40].

Because the heterogeneous nature of MEGO-C's Level-1 base learners makes a single end-to-end explainer prohibitively complex, we implemented a hierarchical two-stage SHAP attribution workflow. In Stage 1, TreeSHAP was applied to the XGBoost meta-learner using only the OOF predictions from a subset of cross-validated, high-skill base learners; ranking their mean absolute Shapley values isolated the most influential contributors. In Stage 2, SHAP was reapplied to this influential subset, but now with the 22 original physical predictors—preclustered into five physically coherent categories (Depth, Optical, Water-intrinsic, Dynamics, and Spacetime; see Table IV)—as inputs, thereby exposing the environmental drivers each model depends upon. This tiered analysis simultaneously reveals the ensemble's internal weighting and the physical mechanisms underpinning its key components, explaining how and why MEGO-C arrives at its predictions.

3) Sensitivity Analysis: We conducted a global Monte-Carlo sensitivity experiment, injecting controlled Gaussian noise into each predictor variable to assess the model's robustness and identify the inputs most critical for accurate cross-dataset transfer [41], [42]. The experiment followed a two-stage design. Stage 1 (one-at-a-time, *OAT*): every non-spatiotemporal feature—coordinates (*JULD*, *LATITUDE*, *LONGITUDE*, *DEPTH*) were deemed error-free—was perturbed independently with zero-mean noise whose variance equaled 5%, 10%, 15%, and 30% of its native variance; the resulting change in RMSE (Δ) quantified that feature's marginal impact. Stage 2 (global interactions): 30 000 Monte-Carlo realizations were generated in which all perturbable predictors were disturbed simultaneously according to the same four-level noise ladder, allowing ΔRMSE to capture synergistic effects. Juxtaposing *OAT* and Monte-Carlo results thus reveals both isolated and joint sensitivities, clarifying which variables dominate error propagation when MEGO-C is applied to datasets with differing signal-to-noise characteristics.

4) Scale-Selective Validation: To confirm that MEGO-C resolves meso- and submesoscale chlorophyll-*a* variability, we applied the FFT-based FlowSieve filter [38] to three dynamically contrasting sectors: 1) the Kuroshio Extension off Japan (140–150 °E, 30–40 °N; area A); 2) the western-Pacific eddy zone (160–170 °E, 30–40 °N; area B); and 3) an equatorial eddy corridor in the northern Indian Ocean (60–70 °E, 10 °S–0 °; area C). For each region, we extracted MEGO-C and benchmark fields (Dataset 2) at the common depths 0.494, 92.326, and 186.126 m, representing the surface, thermocline, and upper-mesopelagic layers of the shared 27-level grid. Notably, all scale-selective patterns shown here are derived from the Chl-a field and should be interpreted as eddy-like coherent Chl-a features, rather than dynamical cyclonic eddies identified from SLA/SSH (e.g., closed contours) or velocity-based diagnostics (e.g., Okubo–Weiss). The resulting depth- and region-specific maps enable direct, one-to-one comparisons that quantitatively and qualitatively assess scale-selective variability and the fine-scale realism of the MEGO-C reconstructions. First, a 10 km low-pass filter was applied to the MEGO-C data to remove potential grid-scale noise. Subsequently, a 25 km high-pass filter was used to isolate the submesoscale component (<25 km) of the signal.

Both datasets were then subjected to complementary 50 km low-pass and 100 km low-pass operations, yielding band-limited fields that retain variance between 50 and 100 km. To quantify scale-selective underestimation in a manner consistent with the targeted scales, we adopted a two-tier feature-masking strategy defined exclusively from MEGO-C. For the large-scale structure, coherent features were delineated from the 100-km low-pass Chl-a field (LP100) and used as fixed control volumes; the benchmark LP100 field was interpolated onto the MEGO-C grid and evaluated within the same masks to compute the within-mask peak Chl-a and area-integrated Chl-a

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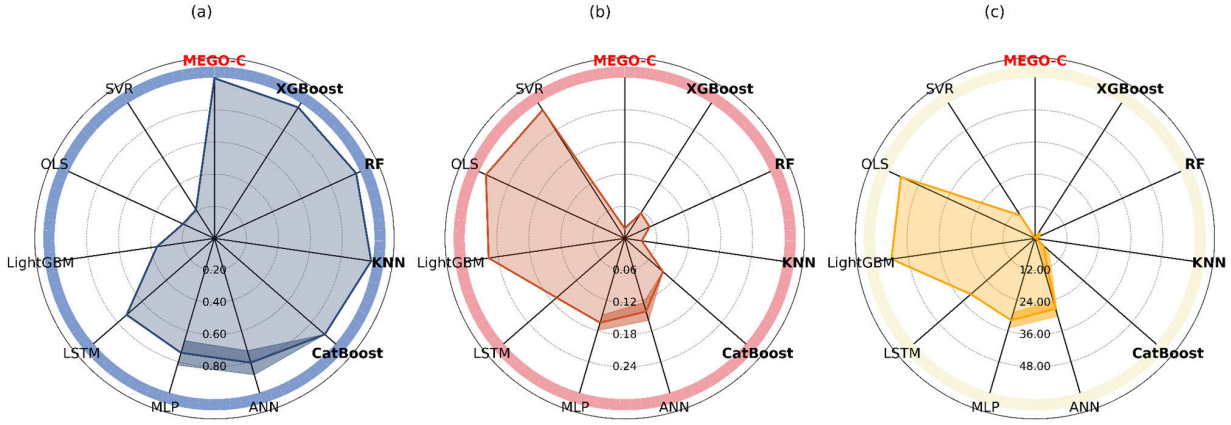


Fig. 3. Performance comparison of base learners and the final MEGO-C ensemble in fivefold cross-validation. Radar plots show (a) R^2, (b) RMSE, and (c) MAPE. MEGO-C (highlighted outer ring) and tree-based models (XGBoost, CatBoost, RF) consistently occupy the high-performance periphery, indicating their superior predictive skill.

($\sum$Chl·dA; a biomass proxy). For mesoscale variability, coherent features were independently delineated from the 50–100 km bandpass field (Band = LP50−LP100), and the benchmark bandpass field was interpolated to the MEGO-C grid and evaluated within the same bandpass masks. Within these masks, we quantified the peak bandpass magnitude (max|Band|) (and, where needed, the area-integrated bandpass magnitude $\sum$|Band|·dA) as a scale-selective measure of biomass-related mesoscale signal. For each metric, percentage underestimation was reported as (MEGO-C − Benchmark)/MEGO-C $\times$100.

The resulting depth- and region-specific maps enable direct, one-to-one comparisons that quantitatively and qualitatively assess eddy content and establish the superior fine-scale realism of the MEGO-C reconstructions.

IV. Result

A. MEGO-C Model Performance

Our rigorous two-step screening procedure conclusively identified tree-based ensembles as the dominant high-performance learners, forming the core of the MEGO-C framework. Step 1—Global skill ranking. Fivefold cross-validation (see Fig. 3) showed that the performance frontier is occupied by tree ensembles—XGBoost, CatBoost, RF—and, to a lesser extent, the distance-based KNN; ANN/MLP/LSTM formed a middle tier, whereas LightGBM, SVR, and ordinary least-squares regression fell well behind. Step 2—Regional dominance check. Spatial "winner-maps" (see Fig. 4) revealed that XGBoost and CatBoost dominate most longitude-latitude-depth bins, with RF, KNN, and the better neural networks (ANN, MLP) emerging as local champions in specific provinces. On the strength of this complementary skill mix, these six learners (XGBoost, CatBoost, RF, KNN, ANN, and MLP) were retained as the Level-1 components that feed the MEGO-C stacking ensemble.

Stacking the top-performing base learners into the final MEGO-C model yielded a decisive performance boost, establishing its clear superiority over any individual algorithm. Fivefold cross-validation (see Fig. 3) already placed MEGO-C at the summit of the skill envelope, but its advantage became unequivocal on the independent test set: the stacked model achieved R^2 = 0.983 with an RMSE = 0.069, handily surpassing the best single learner, XGBoost (R^2 = 0.866, RMSE = 0.195; Fig. 4). Density-scatter validation [see Fig. 5(a)] further shows MEGO-C predictions tightly hugging the 1:1 line, whereas all individual models [see Fig. 5(b)–(k)] scatter more widely—clear evidence that the stacking ensemble synthesizes complementary information from its diverse components to achieve superior generalization across unseen data.

Beyond its headline accuracy, MEGO-C also excels at capturing the detailed shape of vertical chlorophyll-*a* profiles. Fig. 6 compares modeled and observed chlorophyll-*a* profiles from randomly selected BGC-Argo casts, stratified by the depth of the [deep-chlorophyll maximum (DCM), shallow < 20 m; deep > 20 m] [43], [44]. Near the DCM crest, the locality of KNN can occasionally produce the sharpest peak, while tree ensembles track the weak-signal layers below roughly 150 m more faithfully. Across all depths and both profile classes, however, MEGO-C adheres most closely to the in situ curves, confirming that the stacking approach fuses the complementary strengths of distance-based, tree-based, and neural learners and retains top-tier accuracy under contrasting optical conditions.

Finally, to provide an intuitive assessment of MEGO-C's performance, we generated a global map of MAPE on a 1° × 1° grid using the independent validation set (see Fig. 7). Errors stay low across most open-ocean domains—especially within oligotrophic subtropical gyres and the mid-latitude Southern Ocean—where MAPE seldom exceeds 20%. Elevated values (40%–80%) are confined to highly productive regions such as the eastern equatorial Pacific upwelling, the Benguela and Humboldt Currents, and a few river-influenced coastal shelves. The scarcity of bins above 80% highlights the ensemble's overall robustness.

B. Generation and Validation of the Global 3-D Chl-a Product

Prior to the global inversion, we audited and, where necessary, bias-corrected the physical drivers supplied to MEGO-C (Fig. 8, independent-validation subset). The left

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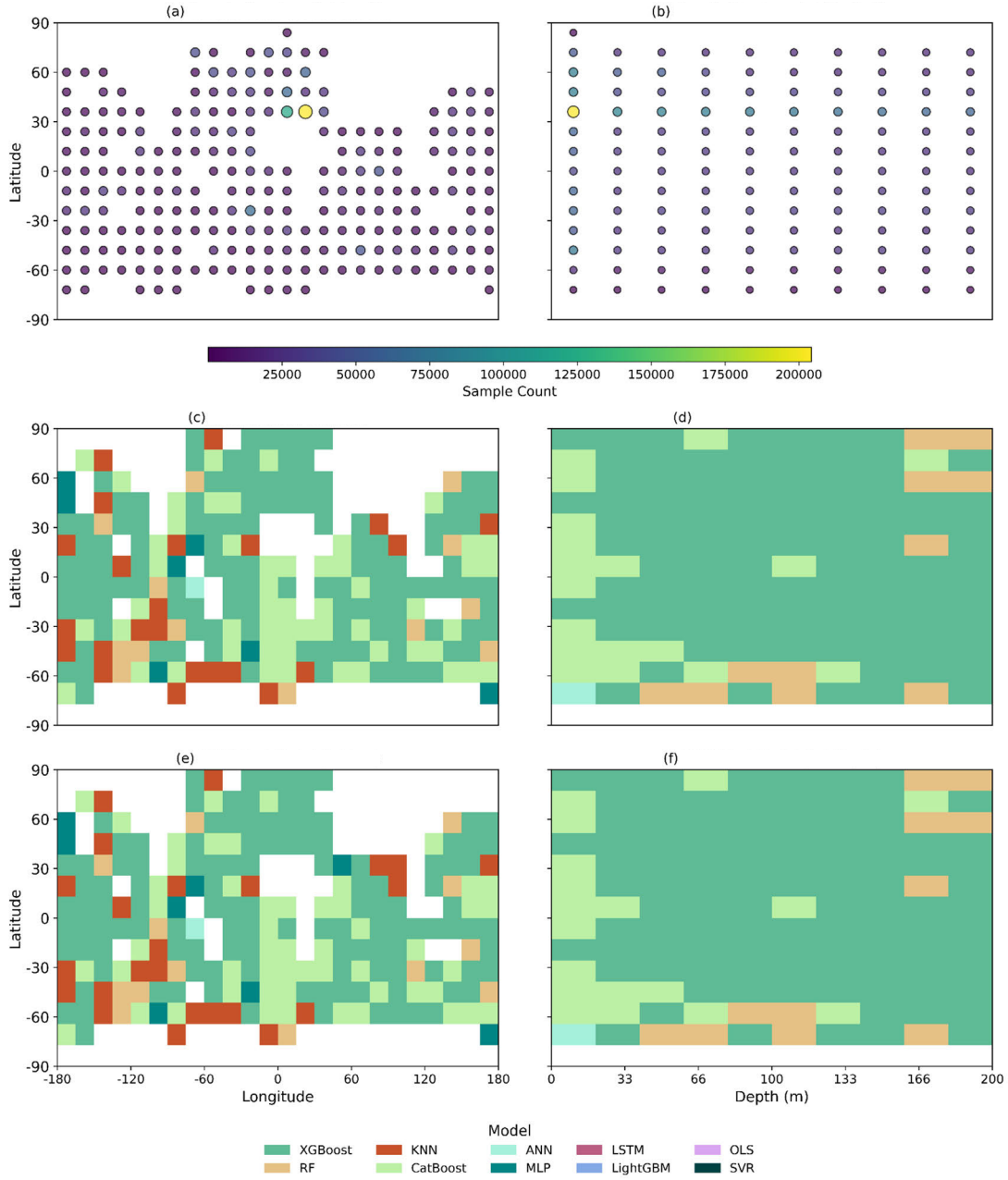


Fig. 4. Spatial dominance analysis of base learner performance. (a) and (b) Sample counts per geographic and depth bin. (c)–(f) Maps showing the best-performing model for each bin based on R^2 and RMSE. XGBoost (dark green) and CatBoost (light green) dominate most regions, justifying their selection as key components of the ensemble. (a) Sample count per lat-lon bin. (b) Sample count per lat-depth bin. (c) R^2 best model (lat-lon). (d) R^2 best model (lat-depth). (e) RMSE best model (lat-lon). (f) RMSE best model (let-depth).

panel confirms that the reanalysis potential temperature (*thetao*) already matches in situ *TEMP* so closely that no adjustment is required. The middle panel, however, exposes a systematic offset between the raw GLORYS12V1 salinity (*SO*) and BGC-Argo *PSAL*, most pronounced in the 10–30 psu range. To remove this bias, an XGBoost regressor was trained with space–time coordinates and *SO* as predictors and *PSAL* as the target, reserving 20% of the match-ups for independent validation. The corrected result (right panel; independent validation) tightens the error envelope—RMSE contracts from 1.642 to 0.169 psu—and eliminates the systematic drift. This step guarantees that the T/S fields ingested by the model are fully consistent with the training data, thereby providing a robust physical basis for the ensuing chlorophyll-*a* reconstruction.

To demonstrate the effectiveness of the salinity correction, we present the correction increment maps $\Delta S = SO_{corrected} - SO_{raw}$ on June 15, 2024, at four representative depths (18.5, 92.3, 155.9, and 186.1 m). As shown in Fig. 9, each panel reports the depth-wise mean and std of ΔSO. Overall, the correction is more pronounced near the surface and rapidly weakens with depth, while the near-zero means indicate a primarily local bias adjustment without introducing a new global offset.

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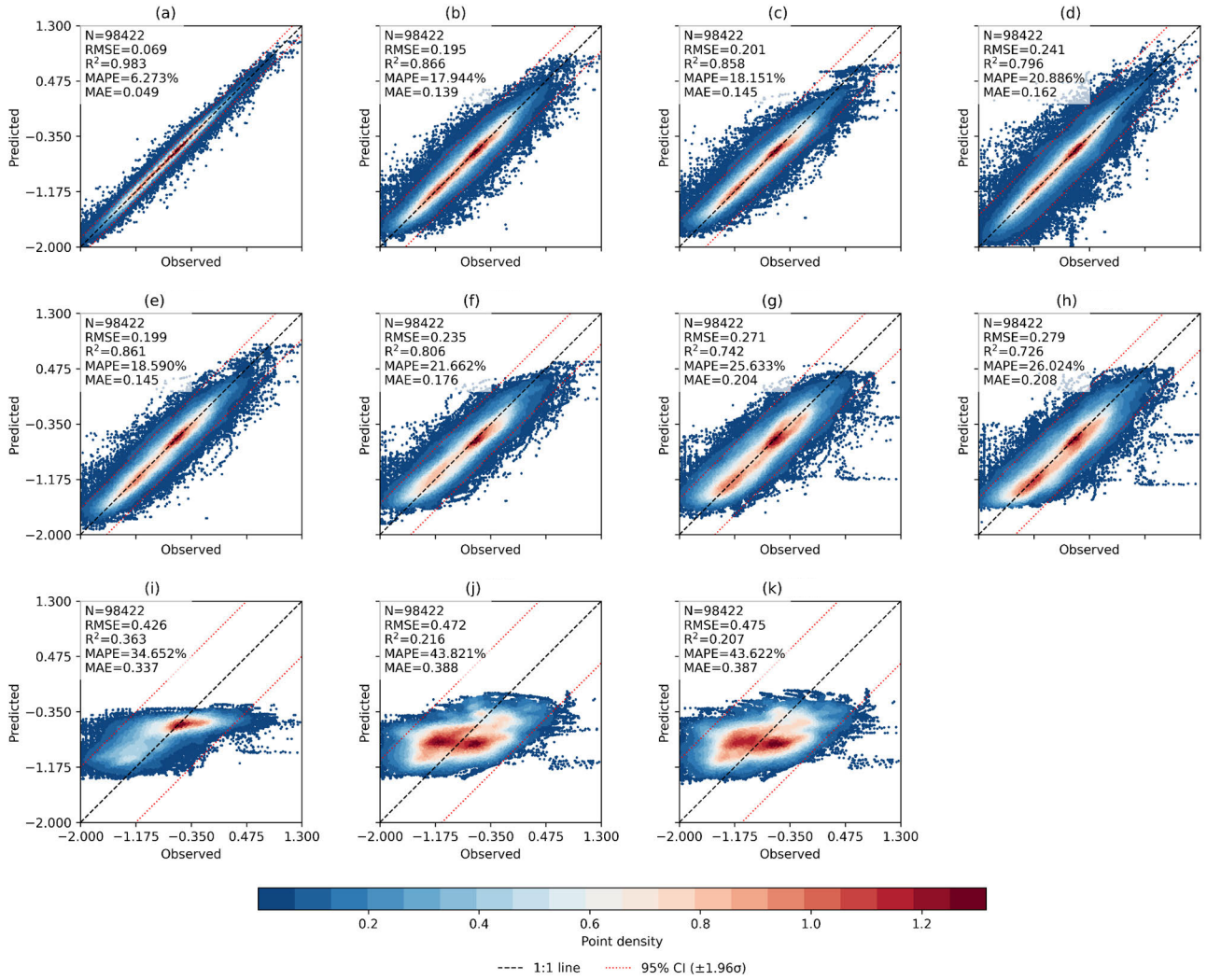


Fig. 5. Validation of MEGO-C and its constituent base learners against the independent test set. The density scatter plot for MEGO-C (a) predictions tightly clustered around the 1:1 line, with the highest R^2 (0.983) and lowest error, confirming (b)–(k) effectiveness of the ensemble approach over individual models.

To assess the reliability of these reconstructions, we applied a three-tier evaluation: 1) a point-by-point statistical comparison of the daily MEGO-C fields and three coarse-resolution products against collocated BGC-Argo measurements; 2) a profile-based check of each dataset's ability to recover key vertical features; and 3) a multiseason, multidepth global visual analysis that tests how well MEGO-C recreates the spatiotemporal evolution of chlorophyll patterns throughout the water column.

To provide an intuitive benchmark of MEGO-C against the reference products, we first collocated each dataset with the BGC-Argo observations in space and time and then carried out a statistical evaluation. When benchmarked against independent BGC-Argo profiles, MEGO-C demonstrated a dramatic improvement in accuracy over existing state-of-the-art products. With an R^2 of 0.891 and an RMSE of 0.191 [see Fig. 10(a)], MEGO-C far surpassed the three CMEMS products, which exhibited R^2 values ranging from just 0.292–0.504 and suffered from significant systematic errors—particularly at depth [see Fig. 10(b)–(d)].

Subsequently, to further compare the profile-level reconstruction skill of MEGO-C with that of the reference datasets, we conducted a month-by-month evaluation against collocated BGC-Argo casts for the overlapping period from November 2021 to November 2022 (see Fig. 11). Because the coarse-resolution reference products do not share identical temporal coverage, restricting the analysis to this common window enables a fair comparison. Rather than presenting 60 separate monthly profile panels, we summarize the diagnostics as time series of both bulk Chl-a skill and DCM structural metrics. Specifically, we first report the monthly R^2 and RMSE of log10(Chl-a) across all paired profiles [see Fig. 11(b) and (c)]. We then extract the DCM characteristics—DCM depth, intensity, and thickness—and quantify their monthly R^2/RMSE [see Fig. 11(d)–(i)]. The DCM is identified as the chlorophyll maximum below 10 m to reduce potential near-surface artifacts, and only profiles with $\geqslant$ 80% valid measurements in the analyzed depth range are retained to ensure robust structure extraction; DCM thickness is defined using a half-maximum criterion (the vertical thickness where

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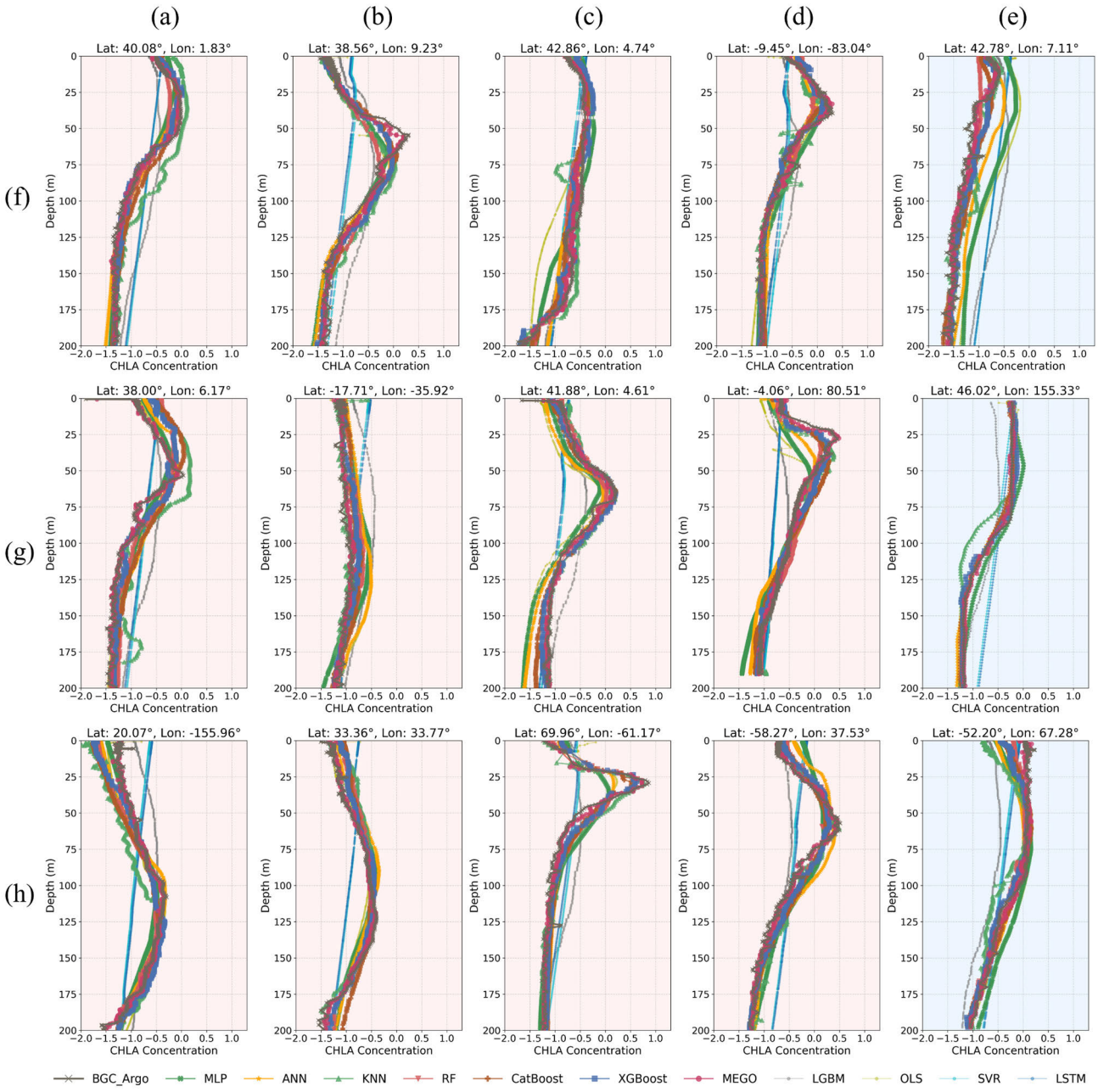


Fig. 6. Vertical profile comparisons for randomly selected BGC-Argo casts. The MEGO-C reconstruction (red line) consistently tracks the in situ BGC-Argo observations (black dots) more closely than any individual base learner across a variety of profile shapes, especially in capturing the depth and magnitude of the DCM. (a)–(h) Each subplot represents a vertical profile comparison for a specific location, with the latitude and longitude given in the title of each individual plot.

Chl-a exceeds 50% of the profile maximum). To make statistical representativeness explicit, Fig. 11(a) reports the month-to-month variation in sample counts, including the total number of paired profiles used for skill evaluation and the subset qualified for DCM diagnostics. Overall, MEGO-C consistently outperforms the reference datasets in both bulk reconstruction accuracy and the fidelity of DCM structure, highlighting its improved physical realism throughout the upper ocean. Finally, to offer a panoramic illustration of MEGO-C's 3-D behavior, we rendered global chlorophyll-*a* maps for February, May, August, and November 2024 at six representative depths—0.5, 47, 92, 131, 186, and 222 m (see Fig. 12). The surface layers reproduce well-known biogeographic contrasts: oligotrophic subtropical gyres (dark blue) flank nutrient-rich coastal and equatorial upwelling belts (orange–red), while the westward-tilting "chlorophyll tongue" in the equatorial Pacific crosses the $\pm 180°$ meridian smoothly, free of edge artifacts. Seasonal transitions are equally clear: austral-summer blooms dominate the Southern Ocean in February, give way to boreal-spring blooms in May, peak in North-Pacific and North-Atlantic storm tracks by August, and re-emerge south of 45 °S in November [45], [46].

With depth, chlorophyll first intensifies and then decays. In the ultra-clear cores of the subtropical gyres, the DCM appears at $\approx$ 130 m (layer 4), whereas in surrounding transition waters it rises to $\approx$ 100 m (layers 2 and 3) [47],

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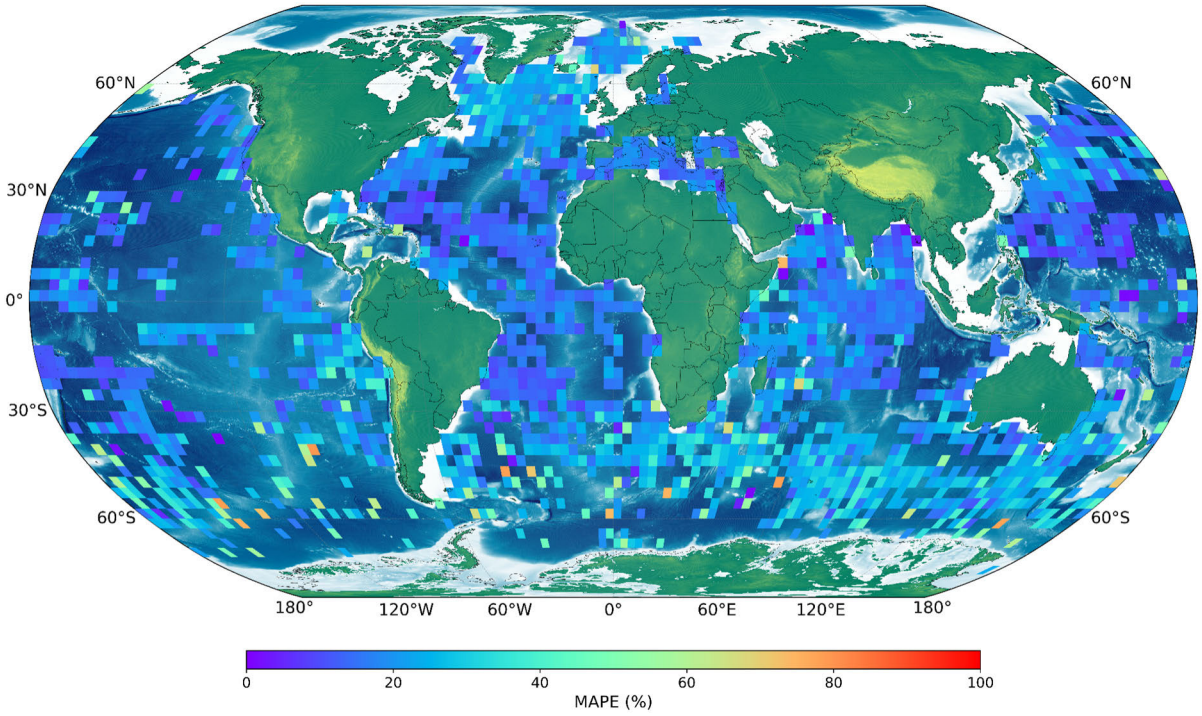


Fig. 7. Global distribution of MEGO-C model error on the independent validation set. The map shows the MAPE calculated on a 1° × 1° grid.

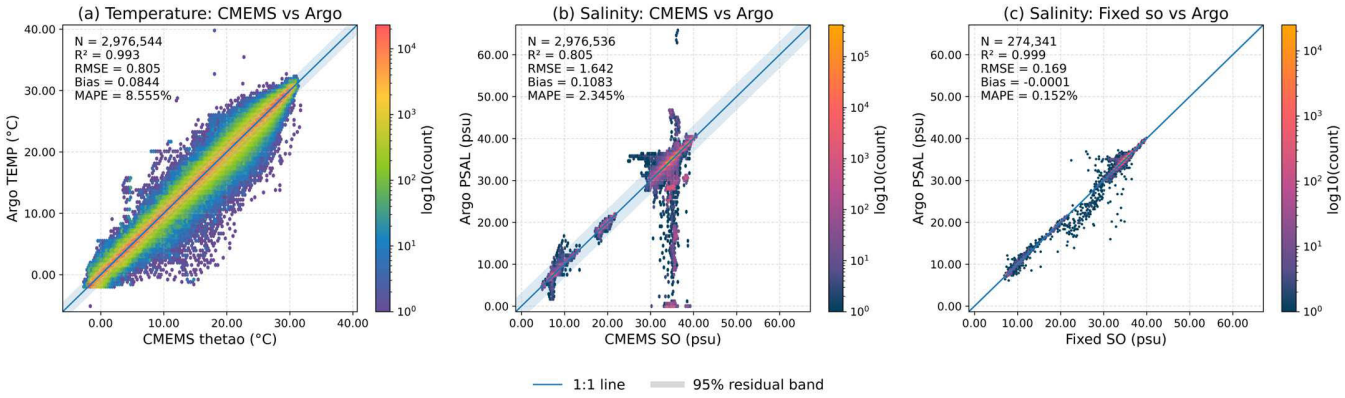


Fig. 8. Consistency check and bias correction of physical drivers. (a) GLORYS12V1 temperature (*thetao*) shows high agreement with BGC-Argo *TEMP*. (b) Uncorrected GLORYS12V1 salinity (*SO*) exhibits a systematic bias against *PSAL*. (c) After bias correction using an XGBoost model (evaluated on an independent BGC-Argo validation set), the corrected salinity field shows near-perfect agreement, ensuring robust inputs for the MEGO-C reconstruction.

reflecting the classic light–nutrient tradeoff. Below ≈ 180 m, the signal fades globally, but residual patches persist beneath major upwelling systems, hinting at particle retention and vertical re-supply [48]. The depth-dependent seasonal pattern is also evident: high-latitude blooms are strong in the upper 50 m yet vanish rapidly below ∼150 m, confirming that MEGO-C captures the vertical attenuation of surface-forced variability [49]. Together, these maps demonstrate that the ensemble not only replicates large-scale geography but also resolves the expected vertical displacement and temporal evolution of the chlorophyll pool across disparate ocean regimes.

C. Resolving Mesoscale and Submesoscale Structures

To quantitatively assess the impact of resolution, we selected three dynamically distinct regions: the Kuroshio Extension, a highly energetic western boundary current system; the western Pacific eddy zone, representative of open-ocean mesoscale eddy fields; and an equatorial corridor in the Indian Ocean, characterized by strong monsoon-driven variability. The ultimate test of MEGO-C's scientific value lies in its ability to resolve the fine-scale oceanographic features that are invisible to other global products. Our scale-selective filtering analysis confirms that MEGO-C not only meets but exceeds this high-resolution promise. The product's native resolution of ∼4 km permits a true scale-aware evaluation of features that are fundamentally absent in coarser products. We first suppressed grid-scale noise with a 10 km Gaussian low-pass filter and then applied a 25 km filter; subtracting the two smoothed fields isolated variability between 10 and 25 km. The resulting bandpass maps [see Fig. 13(b)–(g)] expose a profusion of submesoscale eddies and filaments, particularly along the rims

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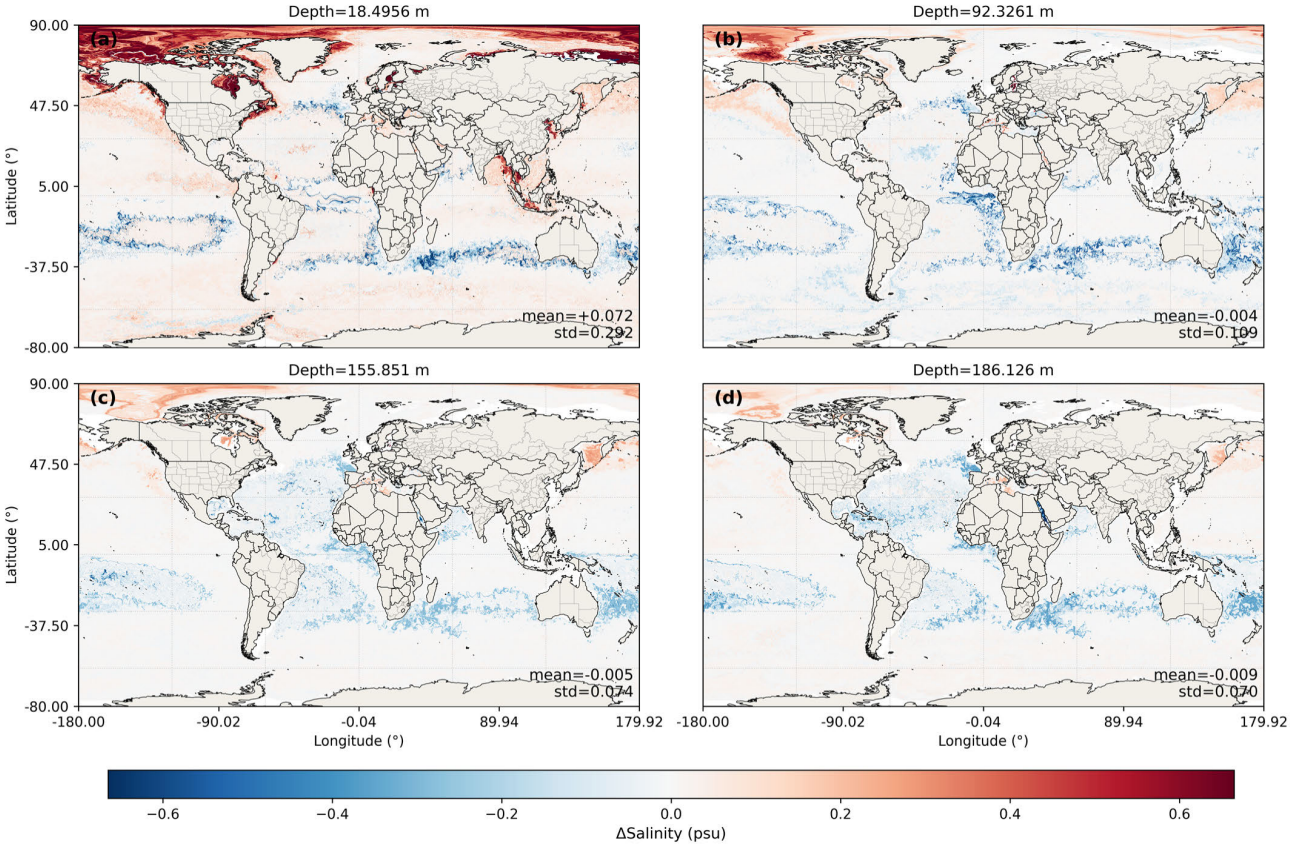


Fig. 9. Spatial distribution of the salinity correction increment, defined as ΔS = so_corrected − so_raw (psu), for the same date (2024-06-15) and depths as in figure (18.50, 92.33, 155.85, and 186.13 m). The difference maps provide a more sensitive visualization of the correction effect than absolute salinity fields; positive (negative) values indicate an increase (decrease) in salinity after correction. The panel-wise mean and standard deviation of ΔS are annotated to summarize the overall bias adjustment at each depth.

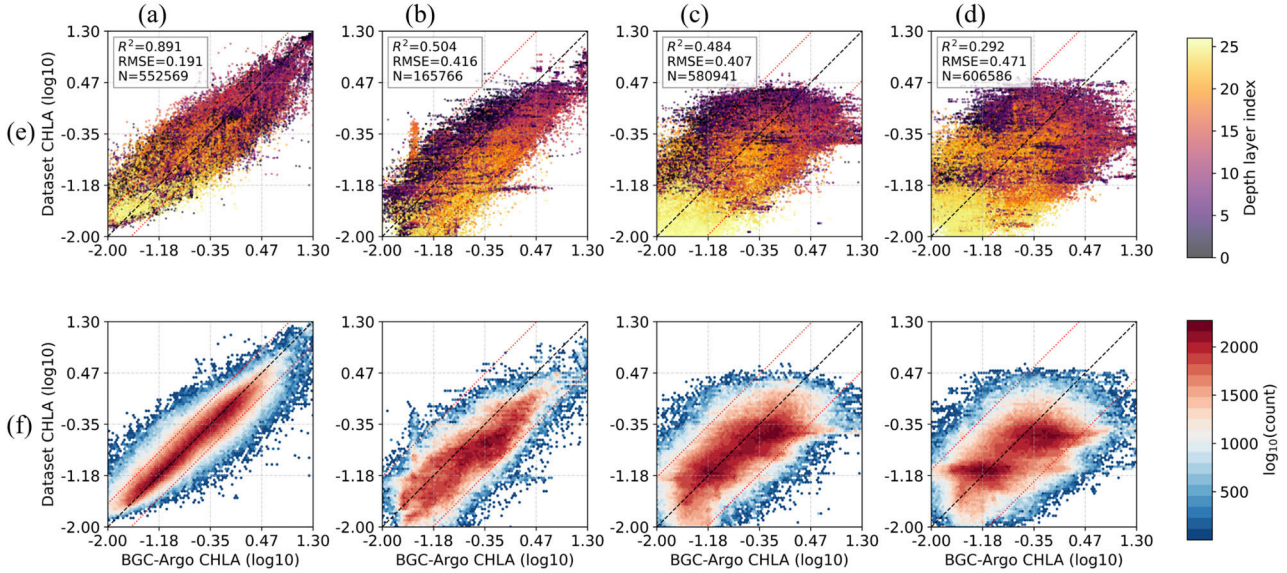


Fig. 10. Head-to-head comparison of MEGO-C and three existing global 3-D *Chl-a* products against in situ BGC-Argo data (July 2021–December 2024). (a) MEGO-C shows the highest correlation (R^2 = 0.891) and the lowest error. (b)–(d) Other products (Dataset 1–3, detailed in Table I) exhibit significantly lower R^2 values and larger systematic biases, as seen by the scattered points and deviations from the 1:1 line. The bottom rows (f) consist of density plots that correspond to the datasets in the columns (e) directly above them. The density plots in the bottom row further highlight MEGO-C's superior accuracy.

of mesoscale vortices and frontal interfaces. Because eddy kinetic energy decays with depth, depth-specific color bars are used, yet abundant 10–25 km structure is visible at all three sampled layers—features that cannot exist in current 0.25° chlorophyll products, whose presmoothing removes <25 km variance.

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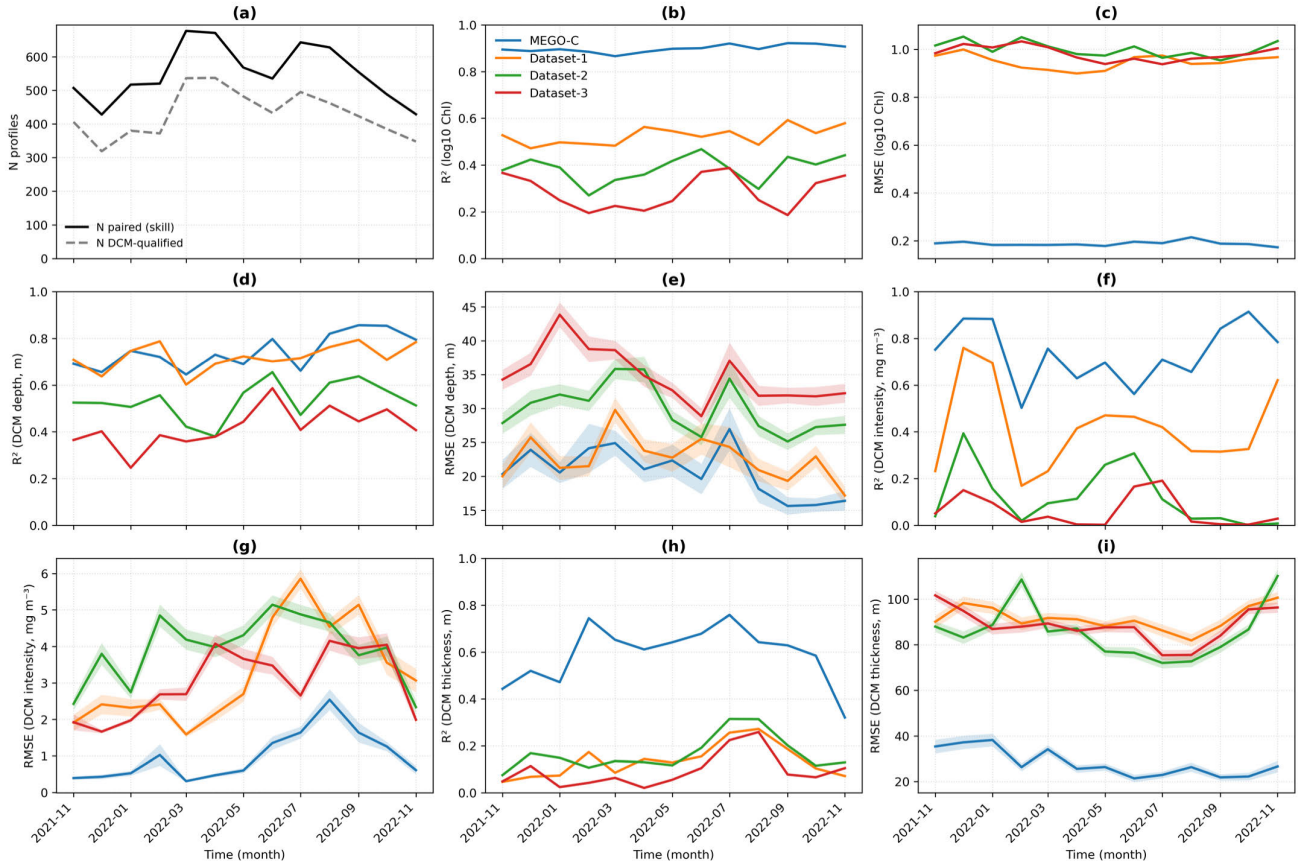


Fig. 11. Monthly skill diagnostics for MEGO-C and benchmark datasets against BGC-Argo (Nov 2021–Nov 2022). (a) Monthly sample counts, including paired profiles used for bulk skill evaluation and the subset qualified for DCM diagnostics. (b) and (c) Monthly R^2 and RMSE of log10(Chl-a) for profile-wise predictions. (d) and (e) Monthly R^2 and RMSE of DCM depth. (f) and (g) Monthly R^2 and RMSE of DCM intensity (mg m^3). (h) and (i) Monthly R^2 and RMSE of DCM thickness (m). The DCM is identified as the chlorophyll maximum below 10 m, and only profiles with ≥ 80% valid measurements in the analyzed depth range are retained for DCM extraction. DCM thickness is defined as the vertical thickness of the layer where Chl-a exceeds 50% of the profile maximum (half-maximum threshold). Shaded envelopes indicate ±1 standard error across profiles.

We then benchmarked MEGO-C against an established 0.25° product (Dataset 1). Both datasets were filtered successively at 50 and 100 km, and their 50–100 km bandpass fields were compared (see Fig. 14). Even at this larger scale, MEGO-C delivers sharper tonal gradients and more coherent eddy cores than Dataset 1, which appears overly smooth and blocky. This superior resolution is not merely a technical achievement; it has profound biogeochemical implications. A case study on a prominent eddy-like Chl-a feature in the Kuroshio Extension region, shown in Fig. 13(a) (surface layer), revealed a critical flaw in current observational capabilities: the coarse-resolution benchmark product underestimated the area-integrated Chl-a ($\sum$Chl·dA; biomass proxy) by 25% and missed the feature's peak Chl-a by 57% in the 100-km low-pass field (LP100). This discrepancy demonstrates that the spatial smoothing in coarse products leads to a significant loss of biomass information within these biogeochemically active features. The loss of detail is even more pronounced when examining the mesoscale variability itself. Here, the 50–100 km bandpass component is defined as Band = LP50−LP100. In the bandpass (50–100 km) filtered anomaly field, the coarse product missed over 98% of the peak bandpass intensity, confirming that the fine-scale structure is almost entirely absent.

Beyond this case study, when we aggregated all identified cyclonic eddies across the three analyzed depth layers and evaluated them in the band-pass domain, the coarse product showed a mean underestimation of 75.25% in peak anomaly and 84.47% in integrated anomaly (both relative to MEGO-C). Together, the single-eddy and multieddy statistics provide convergent evidence that spatial smoothing in coarse-resolution products leads to substantial loss of mesoscale/submesoscale chlorophyll-a structure—and, in the original field, to a significant underestimation of biomass within biogeochemically active features.

Together, these experiments provide clear evidence that MEGO-C simultaneously: 1) retains the submesoscale (<25 km) variability absent from all existing 3-D chlorophyll archives and 2) renders mesoscale (50–100 km) features with higher fidelity. The product is therefore uniquely suited for investigating fine-scale physical–biological interactions in the global 3-D chlorophyll field.

V. Discussion

Our results represent an important step in 3-D chlorophyll-*a* reconstruction; they provide a new lens through which to view the ocean's biological engine. The MEGO-C framework and its resulting atlas make a dual contribution

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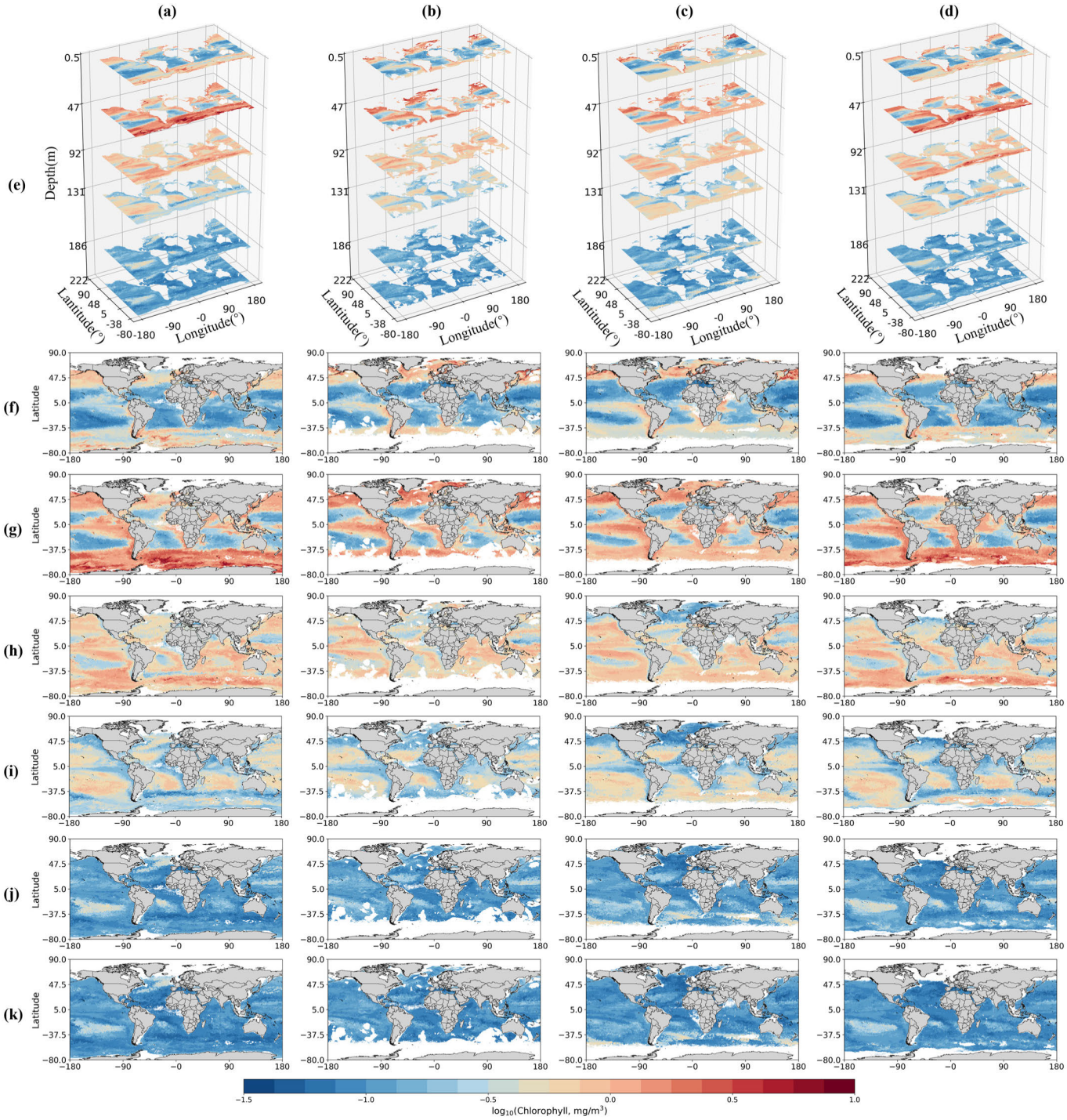


Fig. 12. Global MEGO-C chlorophyll-*a* fields at multiple depths for four representative months in 2024. Columns from left to right (a)–(d) show February, May, August, and November, respectively. For each month, the top panel provides a 3-D stacked perspective, followed by horizontal slices at six depths: 0.5, 47, 92, 131, 186, and 222 mas in (f)–(k). The product reproduces key biogeographic contrasts, resolves the deepening of the DCM in oligotrophic gyres, and captures seasonal bloom dynamics across the upper 220 m.

to the field: they introduce a new paradigm for synergistic data fusion in Earth science, and, by leveraging this new tool, they reveal a systematic and critical underestimation of phytoplankton biomass in the ocean's most dynamic features. This discussion will first explore the profound biogeochemical implications of this finding, then detail the methodological leap that made it possible, and finally, outline the new research frontiers this work opens.

A. New View of Ocean Biogeochemistry: The Potentially Underestimated Impact of Mesoscale Eddies

The ability of MEGO-C to resolve submesoscale features is a valuable advance that provides a new perspective on biogeochemical hotspots in the ocean. Our results provide new observational indications that the contribution of mesoscale eddies to ocean productivity has likely been underestimated by many existing satellite-derived products and models. The

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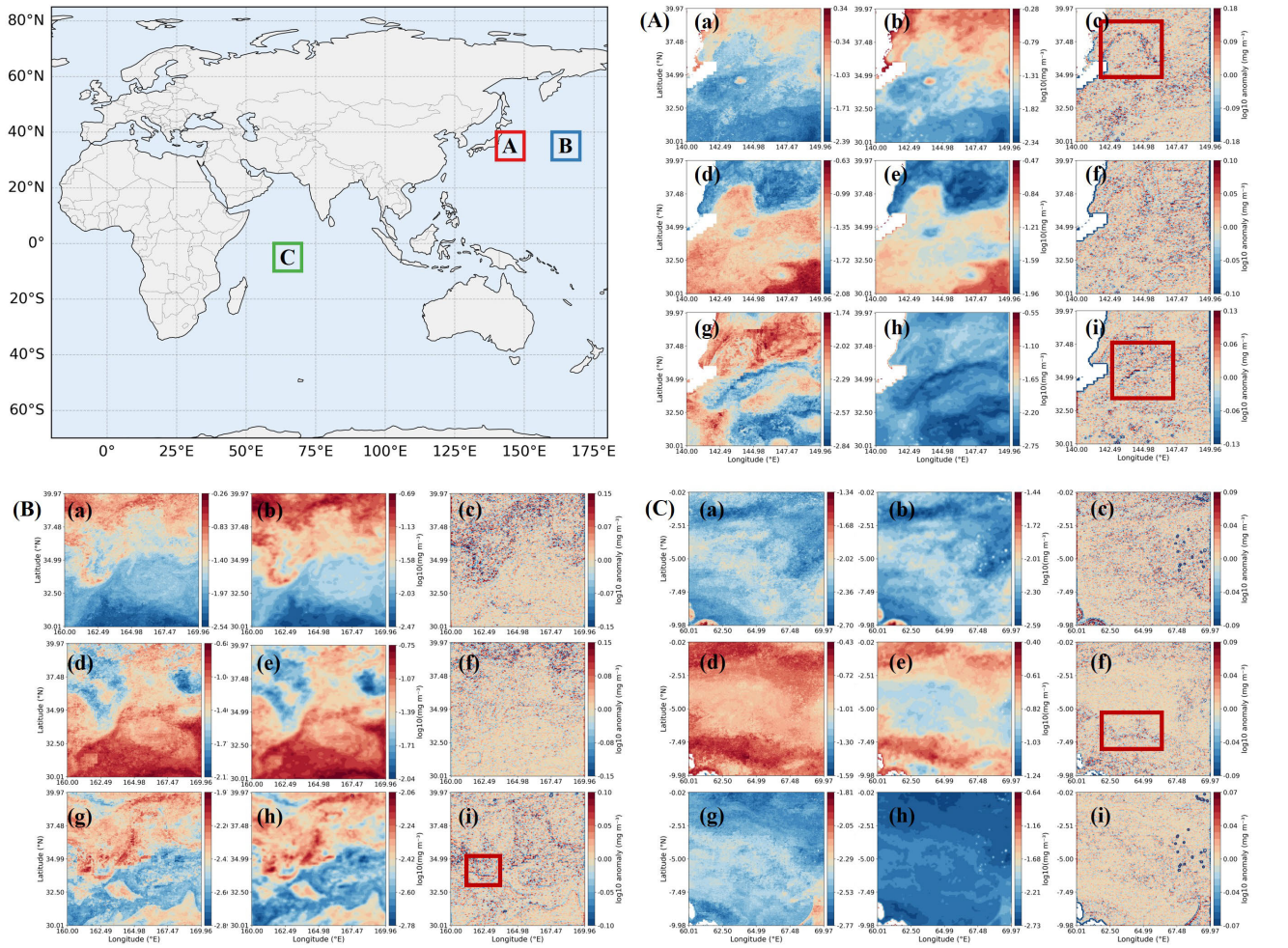


Fig. 13. Submesoscale (10–25 km) structure detection in the MEGO-C chlorophyll-*a* product (April 2024) for three representative regions. (A) Kuroshio Extension, (B) western Pacific eddy zone, and (C) equatorial eddy corridor. For each region, panels are arranged in three rows by depth (top: 0.494 m, middle: 92.326 m, bottom: 186.126 m) and three columns by processing step (left: raw MEGO-C field; center: 25 km low-pass filtered field; right: resulting 10–25 km bandpass field). The bandpass fields reveal rich submesoscale features, including filaments and eddies, which are largely absent in coarser-resolution datasets. Red boxes highlight representative submesoscale structures.

finding that integrated biomass is undervalued by ~25%—and peak concentrations are missed by over 50%—is a notable discrepancy. This suggests that current global carbon budgets, which rely on coarse-resolution inputs, may be missing a significant component of ocean carbon fixation, thereby challenging our fundamental understanding of the efficiency of the biological carbon pump. The discrepancy arises because coarse products average high-concentration filaments and eddy cores with surrounding lower concentration waters, effectively "smearing out" the biological response to the physical forcing. This not only reduces the peak values but also diminishes the total biomass integrated over the feature's area. Given that mesoscale eddies are ubiquitous and are considered key drivers of nutrient injection into the euphotic zone, this underestimation, if representative of a global pattern, could represent a significant missing component in regional and global carbon budgets. A systematic global analysis, now possible with the MEGO-C dataset, is required to confirm this hypothesis. The MEGO-C product, by resolving these features with higher fidelity, provides an essential observational basis for conducting a more accurate global census of the biogeochemical contributions of mesoscale eddies, thereby bridging a critical gap between physical ocean dynamics and marine ecosystem function.

B. Methodological Leap: Interpretable Ensemble Learning for a Complex System

The scientific findings detailed above were enabled by our methodological approach to data fusion. The superior performance of MEGO-C over any single model highlights the power of the stacked, heterogeneous ensemble approach. The vertical distribution of *Chl-a* is governed by a complex, nonlinear interplay of light, nutrients, and physics that varies dramatically across ocean regimes. A single algorithm, whether a tree-based model or a neural network, possesses inherent biases and limitations. By stacking diverse learners, MEGO-C creates a more flexible and robust meta-model that leverages the complementary strengths of its components, leading to lower overall error and better generalization (see Figs. 3 and 5).

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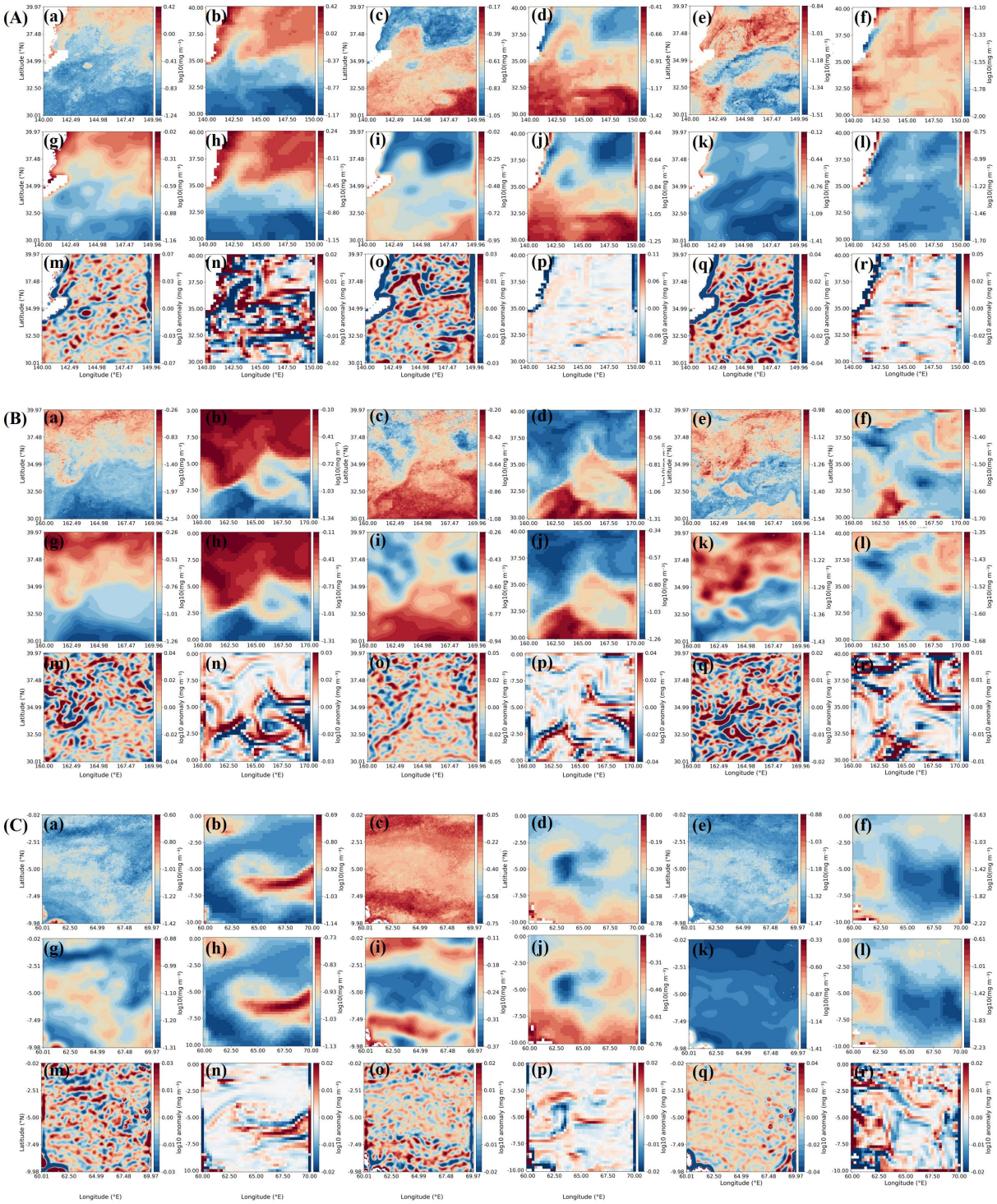


Fig. 14. (A)–(C) (Left of each pair) Comparison of mesoscale (50–100 km) chlorophyll-*a* features between MEGO-C and (right of each pair) a coarse-resolution benchmark product across three regions in April 2024. Data are shown for three depths organized by columns (surface, thermocline, and deep) and three processing steps organized by rows (raw fields, 100 km low-pass, 50–100 km bandpass). The comparison reveals that MEGO-C captures significantly richer and more coherent mesoscale eddy-like Chl-a anomaly structures in the 50–100 km range.

Crucially, this high performance does not come at the cost of interpretability. The SHAP analysis (see Fig. 15) offers a coherent picture of how MEGO-C converts physical and optical inputs into skillful chlorophyll-*a* estimates, confirming its decisions are grounded in established oceanographic principles. The analysis helps to clarify the role of each input parameter. For instance, while some optical variables like the diffuse attenuation coefficient (*Kd*) and Secchi disk depth (*Zsd*) may seem redundant as they both relate to water clarity, the sensitivity analysis (see Figs. 15 and 16) quantifies their distinct contributions to the model's predictive power. Similarly, suspended particulate matter (*SPM*) was included as

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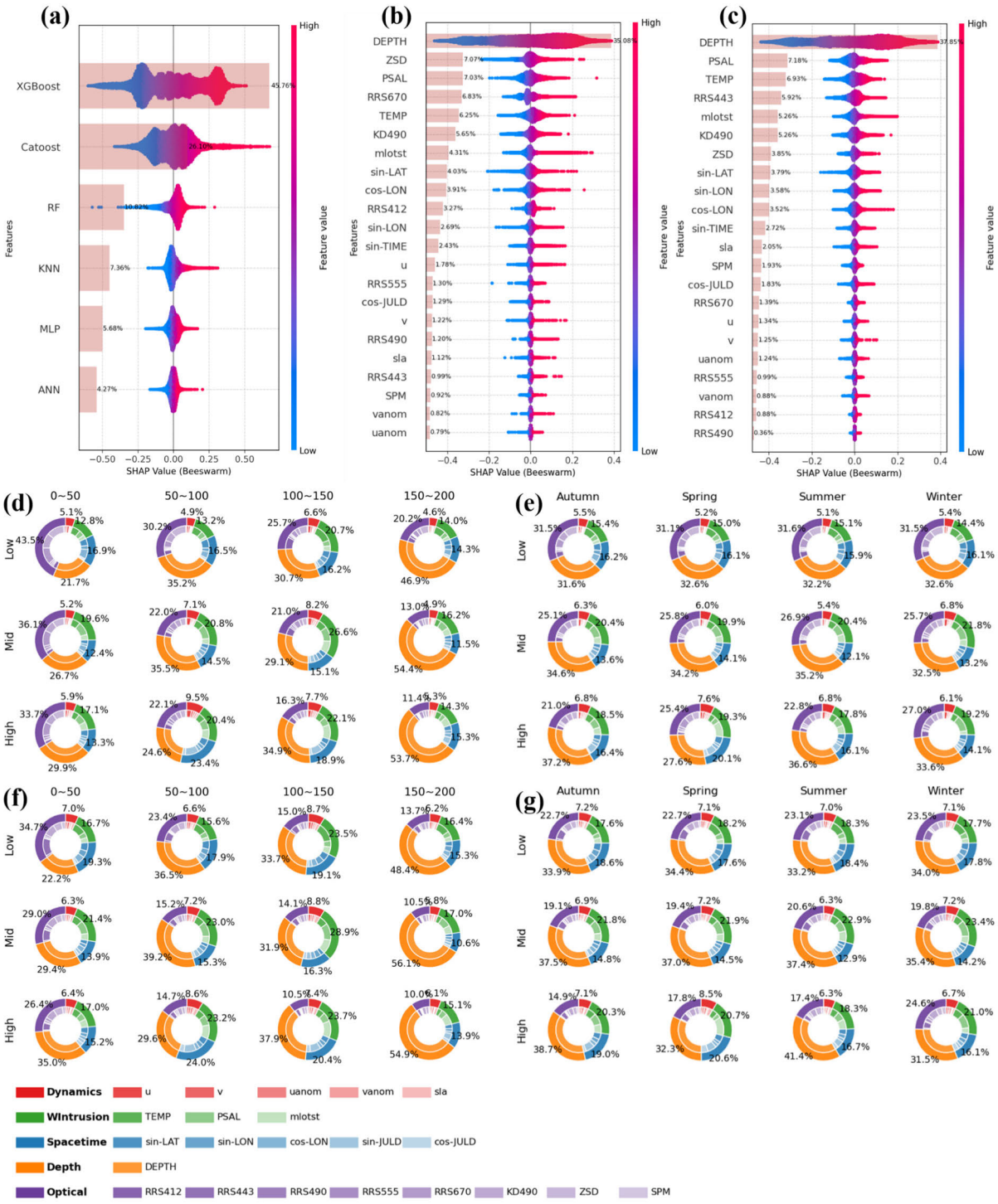


Fig. 15. SHAP-based interpretability analysis of the MEGO-C ensemble. (a) Mean absolute SHAP values for the meta-learner, showing XGBoost and CatBoost as the most influential base models. (b) and (c) Feature importance summaries for the XGBoost and CatBoost base learners, respectively, identifying *DEPTH*, *PSAL*, and *TEMP* as the top predictors. (d)–(g) Grouped SHAP importance stratified by *latitude*, *depth*, and *season*, revealing that the model learns physically consistent relationships.

an input because the model identified it as having predictive value, particularly in optically complex water types, where it provides information not fully captured by other parameters. This data-driven approach, which retains variables based on their demonstrated impact, ensures the model is as robust as possible across diverse oceanic regimes. The analysis reveals that depth, potential temperature, salinity, and mixed-layer depth are the dominant drivers. This ranking aligns with first-principles oceanography: depth encapsulates the compound influence of light attenuation, water-column stratification, and nutrient availability [43], [50]; temperature and salinity reflect water-mass properties and biogeochemical regimes; mlotst

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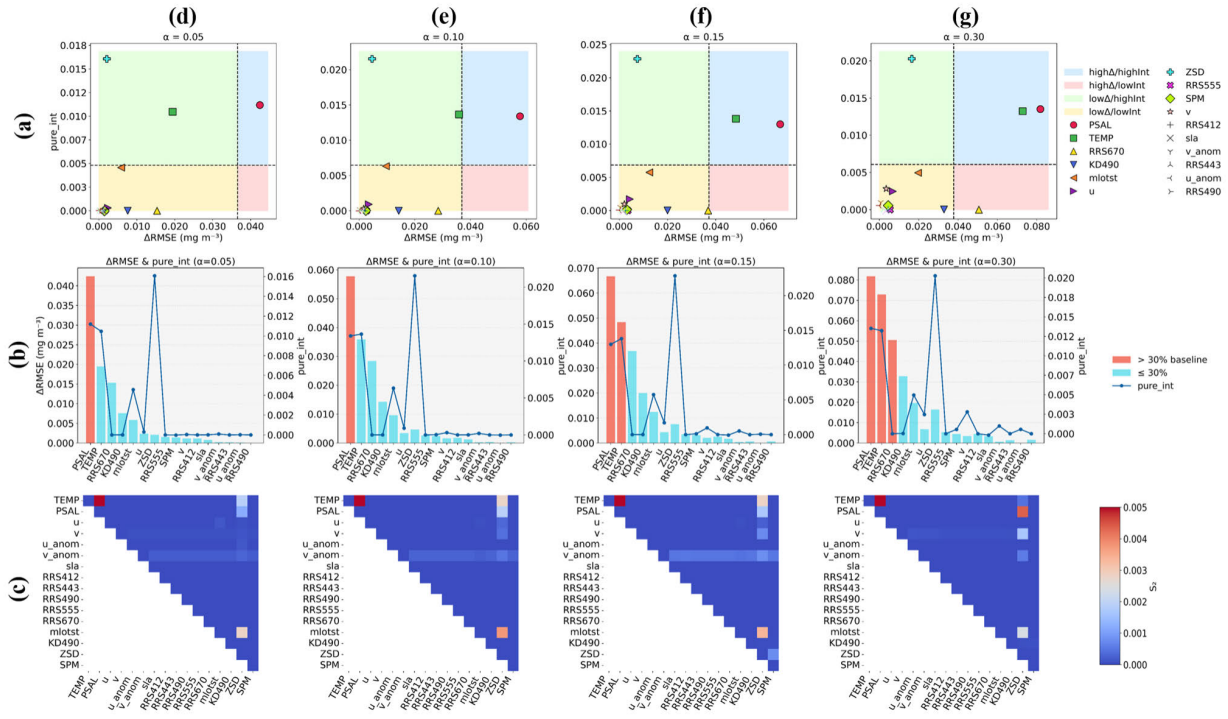


Fig. 16. Monte-Carlo sensitivity analysis of MEGO-C. (a) Feature classification based on direct error contribution (ΔRMSE) and interaction effects at different noise levels. Salinity (PSAL) consistently shows high error and high interaction. (b) Bar charts quantifying the contribution of each feature to model error. (c) Heatmap of second-order Sobol indices, showing that the strongest interaction occurs between temperature and salinity. (d)–(g) Represents a with different values (0.05, 0.10, 0.15, 0.30, respectively).

governs both dilution of surface biomass and entrainment of subsurface nutrient [51], [52]; and optical variables encode key information about phytoplankton pigments, *CDOM*, and light penetration [53], [54], [55]. Notably, our data-driven findings align remarkably well with fundamental ecological theory. The high importance assigned to depth and mixed-layer depth directly reflects the physical constraints on phytoplankton growth described by Sverdrup's critical depth theory [56]. Similarly, the model's reliance on temperature and salinity as proxies for water masses implicitly captures the large-scale patterns of nutrient availability that govern phytoplankton biogeography. This convergence between our machine learning model's learned relationships and established ecological principles provides strong evidence that MEGO-C is not merely fitting data but is capturing the underlying biogeochemical mechanisms.

To test whether these relationships remain physically sensible across space and time, the 22 original predictors were grouped into five coherent classes (Dynamics, Water-intrinsic, Spacetime, Depth, and Optical), SHAP values were recomputed at the group level, and samples were binned by latitude × season and latitude × depth [see Fig. 15(d)–(g)]. The grouped-SHAP results [see Fig. 15(d)–(g)] show that MEGO-C's two leading learners—XGBoost and CatBoost—draw on the same, physically coherent set of predictors.

In latitude-depth space [see Fig. 15(d) and (f)], depth is already influential in the upper 50 m and becomes the single largest contributor below about 100 m, consistent with the growing control of light attenuation and stratification with depth. Optical variables dominate the surface layers at all latitudes but lose importance rapidly beneath ≈ 100 m once irradiance is strongly diminished; their share is greatest in optically clear, low-latitude waters. Toward high latitudes, the relative importance of Optical predictors decreases because chlorophyll variability is more strongly governed by mixed-layer dynamics and nutrient replenishment (e.g., deep winter mixing and episodic restratification), reducing the marginal explanatory power of instantaneous surface optical signals. In addition, lower solar elevation, more complex nonalgal optical constituents (CDOM/NAP) [57], and higher atmospheric-correction uncertainty can further weaken the unique information content of surface optical variables [58], [59], [60]. Notably, this pattern is not an artifact of gap filling, since we do not interpolate cloud-missing optical inputs prior to inversion; SHAP is computed conditional on available observations, although persistent cloudiness may indirectly reduce effective sampling and increase noise in high-latitude optical measurements. Water-intrinsic factors (*TEMP*, *PSAL*, and *-mlotst*) peak between 50 and 150 m in every latitude band, where nutrient entrainment and remineralization shape chlorophyll gradients. This mid-depth maximum is physically expected because the 50–150 m layer typically spans the transition from the mixed-layer base into the upper thermocline/pycnocline [61], where T/S gradients strengthen and the upper nutricline is frequently encountered [62]. In this depth range, modest changes in stratification and mixed-layer depth can strongly modulate vertical nutrient supply and the depth/intensity of the DCM [43], [50], which explains why TEMP/PSAL/mlotst carry elevated explanatory power [63]; the exact peak depth may shift across regions and seasons

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as the thermocline/nutricline structure varies. By contrast, Spacetime coordinates and surface Dynamics together account for less than 15% of the explanatory weight in most bins, rising only slightly at high latitudes or in the shallowest layer—evidence that large-scale geography and currents provide secondary, not primary, information. The latitude-season view [see Fig. 15(e) and (g)] reveals that within a given latitude belt, the relative weights change little from autumn through summer, confirming that the model does not hinge on a single season. Optical cues remain the top drivers in the tropics, whereas water-intrinsic variables and depth gain importance toward mid- and high latitudes, in line with stronger thermocline control outside the tropics. Spacetime indices contribute modestly everywhere, furnishing a seasonal phase reference without overwhelming the physically driven terms.

Taken together, these depth- and season-dependent patterns accord with first-principles ocean optics and mixed-layer physics, demonstrating that the dominant learning pathways extract complementary, meaningful information and that the ensemble remains robust and interpretable across latitude bands, depths, and seasons.

C. Model Robustness and a Framework for Future Data Products

The Monte-Carlo sensitivity experiment (see Fig. 16) augments the SHAP diagnosis by providing clear, actionable guidance on the model's robustness. The results confirm that ensuring bias-free salinity and temperature fields is paramount when forcing MEGO-C with alternative datasets. Our preemptive bias correction of GLORYS12V1 salinity illustrates the type of rigorous preprocessing that safeguards model performance.

From a broader perspective, the MEGO-C framework contributes to the practice of geophysical data product generation. Our work demonstrates a systematic approach that explicitly accounts for inter-source discrepancies and uncertainties. The comprehensive Monte-Carlo analysis provides a quantitative assessment of how uncertainties propagate through the model, offering crucial guidance for both end-users and developers. Finally, by employing SHAP, we enhance the transparency and trustworthiness of a complex machine learning model in a geoscience application, moving beyond a "black box" approach. This integrated methodology of bias correction, uncertainty quantification, and model interpretability presents a robust framework for developing next-generation remote sensing data products.

These results align with the SHAP importance ranking and have immediate operational implications. When MEGO-C is forced by alternative reanalyses, climate-model output, or future satellite products, ensuring bias-free salinity and temperature fields is paramount. Our preemptive bias correction of GLORYS12V1 salinity (see Fig. 8) illustrates the type of preprocessing that safeguards model performance and should be replicated whenever upstream data deviate from the training climatology.

From a broader perspective, the MEGO-C framework contributes to the practice of geophysical data product generation. Our work demonstrates a systematic approach that explicitly accounts for intersource discrepancies and uncertainties. The comprehensive Monte-Carlo analysis provides a quantitative assessment of how uncertainties propagate through the model, offering crucial guidance for both end-users and developers. Finally, by employing SHAP, we enhance the transparency and trustworthiness of a complex machine learning model in a geoscience application, moving beyond a "black box" approach. This integrated methodology of bias correction, uncertainty quantification, and model interpretability presents a robust blueprint for developing next-generation remote sensing data products.

D. Limitations and Future Frontiers

The primary limitations of the current product are intrinsically linked to the global observing system itself. First, its accuracy is fundamentally constrained by the quality of the input fields. Even after bias correction, the product's fidelity still hinges on the realism of GLORYS12V1 T/S data and Copernicus ocean-color observations. Second, the BGC-Argo float array, though extensive, remains sparse in certain regions—particularly marginal seas and high latitudes—so local skill may vary. Performance may also drop in optically complex coastal waters, where satellite retrievals are inherently more uncertain.

Future work should therefore pursue several avenues. Adding new predictors—such as nitrate fields from biogeochemical models or remote-sensing proxies—could further raise accuracy. As the BGC-Argo archive expands, periodic retraining will refine regional skill and shrink uncertainties. Furthermore, the advent of new satellite missions will provide exciting opportunities for validation and refinement. For example, data from the Surface Water and Ocean Topography (*SWOT*) mission could offer an independent, high-resolution dataset of sea surface height and fine-scale ocean features, providing a valuable means to confirm and further investigate the eddy dynamics revealed by MEGO-C. Finally, deploying the MEGO-C dataset in applied studies—for instance, to quantify the contribution of submesoscale eddies to regional and global carbon export, to investigate how climate-driven changes in ocean stratification are altering the vertical structure of phytoplankton communities, or to constrain high-resolution primary production estimates—will provide the ultimate test of its scientific value.

Moreover, because daily satellite optical predictors are frequently affected by cloud coverage, we do not perform spatiotemporal gap filling on these inputs prior to inversion. MEGO-C is generated only where all required surface predictors pass quality control; otherwise, the corresponding grid cells are retained as missing (NaN) in the 3-D product. To further improve spatiotemporal coverage in persistently cloudy regions, we plan to investigate postprocessing strategies that operate directly on the reconstructed 3-D fields by exploiting 3-D spatial coherence and temporal evolution, thereby avoiding multistage uncertainty propagation and potential smoothing artifacts introduced at the 2-D surface-input level.

VI. Conclusion

In this study, we have addressed a long-standing challenge in Earth observation by developing and validating MEGO-C,

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a new global, daily, submesoscale-resolving 3-D chlorophyll-a atlas. The key contributions and their implications are as follows.

1) *A New Observational Window Enabled by a Novel Framework:* We have introduced a powerful, transparent, and transferable data fusion framework (MEGO-C). Its application provides the first-ever global, 4-D view of ocean chlorophyll at the submesoscale. This reveals a systematic underestimation of biomass in mesoscale eddies, opening new frontiers for studying the fine-scale drivers of marine productivity and challenging current assumptions in global biogeochemical models.

2) *A High-Fidelity Benchmark for Earth System Models:* With its high demonstrated accuracy (R^2 = 0.89), unprecedented resolution, and physical consistency, the MEGO-C atlas serves as a valuable new benchmark for evaluating and improving the biogeochemical components of global climate and Earth system models. It provides a much-needed observational constraint on fine-scale processes that these models are just beginning to resolve.

3) *A Transferable, "White-Box" Approach for Data Fusion:* The MEGO-C framework, with its emphasis on heterogeneous ensembles and "white-box" interpretability (via SHAP analysis), offers a robust and transferable approach for the 3-D reconstruction of other essential ocean and climate variables, such as ocean salinity, temperature, or dissolved oxygen. This demonstrates a pathway for developing the next generation of intelligent, trustworthy, and physically grounded geophysical data products.

In summary, MEGO-C does not merely fill a data gap; it provides the community with a new observational tool that represents a paradigm shift in our ability to observe ocean biology. It transitions our view from a coarse, 2-D surface to a high-resolution, 3-D volume, encouraging the scientific community to re-evaluate the fine-scale drivers of global biogeochemical cycles and providing an essential resource for the next generation of research into the ocean's role in the Earth system.

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